SATA 1D: How It Works, Verification and Results

Position-dependent correction of the topographic bulk residual ΔC_0
in multichannel azimuth reconstruction — systematic verification of the implementation

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September 2026

Abstract

This report documents the one-dimensional SATA correction of the bulk topographic residual ΔC_0 implemented in the `sata1d` package, and a systematic verification of that implementation. Section 2 states the algorithm in one page: for every channel, a map $\Delta C_0[x]$ is written on the azimuth cells where each scatterer's energy appears in the sub-aperture spectra (Stage A), and a sub-aperture loop maps every Doppler bin through the beam angle to a cell, reads the map there and rotates the bin by $e^{+j2\pi\Delta C_0/\lambda}$ (Stage B). Sections 3 and 4 derive each equation from the geometry in the order the code executes it, next to the code. Section 6 verifies the implementation in three layers: a review of every sign and index convention against the geometry and the reconstruction; twenty tests, from the algebra of the kernel to reconstructions of multi-target scenes compared with an oracle that corrects each scatterer with its own residual; and the geometry sweep of the case matrix (181 scenes) plus the topographic ramp (18 scenes). All pass/fail tests pass. For one target the azimuth ambiguity returns from $-23\ldots-9$ dB to -39.4 dB, the level of the reference, at every height up to 400 m; only the removal sign reaches it (the opposite sign gives -7.8 dB). With two targets 3 km apart each reads its own residual to 2 %. Two findings are documented. The per-sub-band kernel used the axis $f_k + \text{fftfreq}$ with positions measured from β_k; it is replaced by the absolute, f_k-centred axis on the broadside grid prescribed in Section 4.2 of the report. And a limit of any per-channel correction on the aliased line was identified and measured, the Doppler-alias collision: two scatterers whose azimuth separation is a multiple of the fold distance $X = 6.2$ km, to within the main-lobe width of the sub-aperture response (0.6 km), share the same sub-aperture bins in every window, and no bin-wise correction can serve both. Testing it exposed a weakness of the map for separations near $X \pm h_w$, which was repaired (the nearest image now wins where images overlap); outside the collision core SATA equals the oracle, inside it the ambiguity still improves by 10–13 dB but stays 13–16 dB above the oracle. On the sweep, the new map improves the median ambiguity by 3.7 dB over the previous one and holds the oracle level as long as neighbouring targets differ by less than about 100 m in height.

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Notation

Symbol	Meaning	Value in the examples
λ, v	wavelength, sensor velocity	0.25 m, 7688.5 m/s
r, r_0	reference slant range (ruler and δx)	766.2 km
θ	look angle at the reference point	20.0°
PRF, $\mathrm{PRF}_{\mathrm{ch}}$	system PRF, PRF of one channel (= PRF/N_{rx})	2000 Hz, 500 Hz
N_{rx}	number of receive channels	4
$b_{at,i}$, $b_{xt,i}$	along-/cross-track baseline of channel i	7.69 m steps; $[63.7, 27.0, 4.1, 1.7]$ m
$d_s = v/\mathrm{PRF}_{\mathrm{ch}}$	azimuth cell of one channel	15.38 m
$N_{a,ch}$	samples per channel line	3072
δx	SATA (sub-aperture) resolution	309.5 m
$T_{\mathrm{sub,eff}}$, hop, N_{zp}	sub-aperture length, hop, FFT length	20, 10, 128 samples
f, ϕ	bin frequency (absolute label), baseband bin frequency	
$\beta(f)$	beam (squint) angle belonging to f	
$x(f)$	cell offset of bin f from the window centre (the *ruler*)	0.81 cells/Hz
c_w	centre cell of window w	
x_t	cell of a scatterer on the channel line	1537 (channel 0)
h_w	half-width of a scatterer's footprint in the map	82 cells (1.26 km)
X	cell shift of one $\mathrm{PRF}_{\mathrm{ch}}$ fold	405.06 cells (6.23 km)
B	Doppler bandwidth of a target	1281 Hz
f_k, β_k	centre frequency and squint of output sub-band k	± 250, ± 750 Hz
$C_{0,i}(p)$	bulk coefficient of channel i for point p	
$\Delta C_{0,i}(p)$	residual $C_{0,i}(p) - C_{0,i}(p_{\mathrm{ref}})$	-5.47 cm (ch. 0, $h = 240$ m)
φ_i	residual phase $2\pi\Delta C_{0,i}/\lambda$	$-78.8°$ (ch. 0, $h = 240$ m)

1 What SATA Corrects

The residual. The reconstruction filters of channel i are built from the coefficients of the reference point p_{ref}, the point on the reference surface at the same slant range and azimuth (Fig. 1). The transfer function of channel i carries the bulk term $e^{-j2\pi C_{0,i}/\lambda}$ (`filter_matrix` in `reconstruction.py`), so a scatterer p at height h leaves in the data the factor

$$e^{-j2\pi C_{0,i}(p)/\lambda} = e^{-j2\pi C_{0,i}(p_{\mathrm{ref}})/\lambda}\, e^{-j\varphi_i(p)},$$
$$\varphi_i(p) = \frac{2\pi}{\lambda}\,\Delta C_{0,i}(p), \qquad \Delta C_{0,i}(p) = C_{0,i}(p) - C_{0,i}(p_{\mathrm{ref}}), \tag{1}$$

of which the filter removes only the first factor. For small heights the residual has the closed form of the interferometric height sensitivity (report, Eq. 37),

$$\Delta C_{0,i} \simeq -\frac{b_{xt,i}\,\Delta h}{r_0 \tan\theta}. \tag{2}$$

For channel 0 ($b_{xt} = 63.7\,\mathrm{m}$) and $h = 240\,\mathrm{m}$, Eq. (2) gives $-5.47\,\mathrm{cm}$, a phase of $\varphi_0 = -78.8°$; for channel 3 ($b_{xt} = 1.7\,\mathrm{m}$) it is $-0.14\,\mathrm{cm}$ ($-2.0°$).

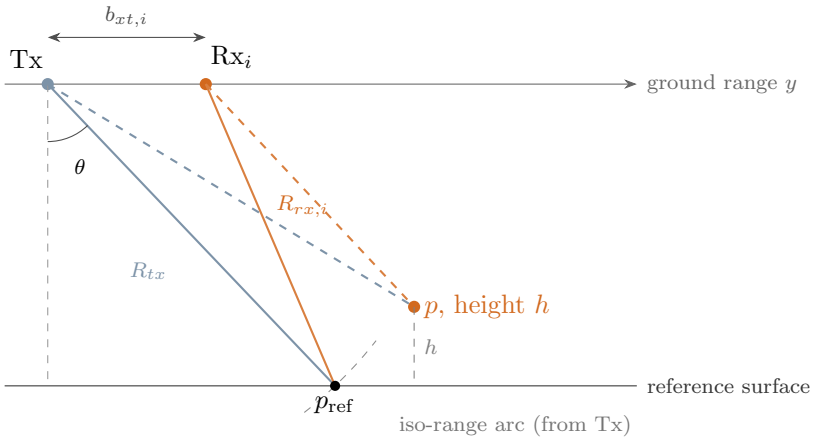


Figure 1: What is corrected. The reconstruction filter is built for p_{ref} (solid paths). The actual scatterer p lies at the same slant range from the transmitter but at height h, further out on the iso-range arc (dashed paths). The two positions are equivalent for the transmitter but not for a receiver displaced by $b_{xt,i}$: the bistatic range differs by $\Delta C_{0,i}$, which depends on the channel.

Why it matters, and why one multiplication is not enough. A residual common to all channels costs nothing: it multiplies the whole image by one phase (report, Eq. 40). What degrades the reconstruction is the *spread* of φ_i across channels (report, Eqs. 41–42): the unmixing of the aliased sub-bands is then incomplete and the missing fraction reappears as azimuth ambiguities. If the scene had a single height, multiplying channel i by $e^{+j\varphi_i}$ would remove the residual exactly. But φ_i depends on the scatterer, and every channel sample receives echoes from every scatterer the beam illuminates at that instant. The phase to remove therefore has to be selected *per scatterer*, without focusing the line. That is what SATA does.

2 SATA 1D at a Glance

The principle in one sentence. In a short sub-aperture the platform hardly moves, so a scatterer's echo is a tone whose Doppler frequency f tells where the scatterer is: $f = (2v/\lambda)\sin\beta$, and the beam angle β points at its position on the ground. The Doppler bins of a sub-aperture are therefore *positions*, and each bin can be rotated by the residual of the position it looks at.

The two stages. SATA acts on each channel line separately, before the reconstruction, and always in the same order (Fig. 2). Stage A builds the map $\Delta C_0[x]$: for every scatterer, its residual $\Delta C_{0,i}$ is written on the cells where its energy can appear in the sub-aperture spectra, its own cell $x_t \pm h_w$ and the folded copies at $x_t \pm X$; zero elsewhere. Stage B runs the sub-aperture loop: cut, weight, FFT, map every bin to a cell through the ruler $x(f)$, read the map there, rotate, inverse FFT, overlap-add.

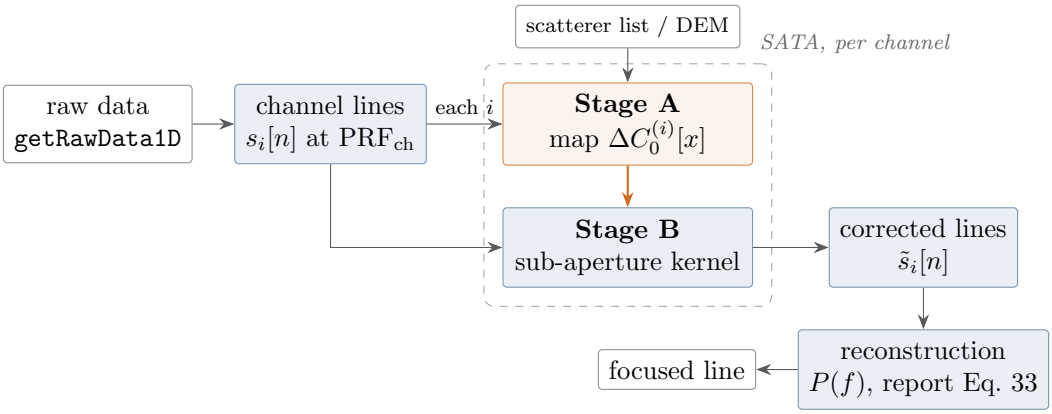


Figure 2: Place of SATA in the chain. Stage A needs the scatterers (in simulation, the scene list; with real data, a DEM). Stage B reads the map and rotates the Doppler bins. Whole band: one corrected set feeds all sub-bands of the reconstruction. Per sub-band: both stages run once per output sub-band k (Section 5).

One window on real data. Figure 3 follows a single sub-aperture window of channel 0 through Stage B, for the target at $h = 240\,\mathrm{m}$ (cell 1537). The window is centred 70 cells before the target. Its spectrum peaks at $f = +86\,\mathrm{Hz}$, which the ruler (0.81 cells/Hz) turns into +70 cells: the bin points at cell 1537, the target's own cell. The map holds $-5.47\,\mathrm{cm}$ there, so the peak is rotated by $+2\pi(-0.0547)/0.25 = -78.8°$, exactly the conjugate of the residual phase of Eq. (1). Bins that point outside the target's block read zero and are left untouched. Every window does the same, and the corrected line equals the ideally rotated one to 2 %.

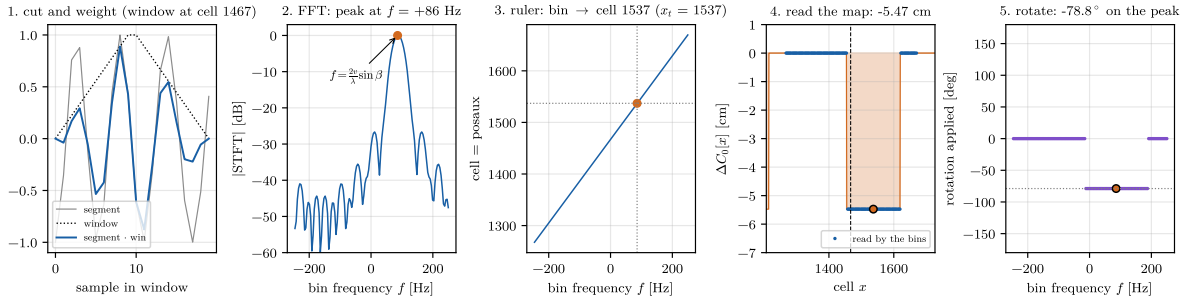


Figure 3: Stage B on one window (channel 0, window centred at cell 1467, target at cell 1537). 1: the segment and the triangular weight. 2: its spectrum, peak at +86 Hz. 3: the ruler maps the peak bin to cell 1537. 4: the map is read at the cells the bins point to (blue dots); the peak reads $-5.47\,\mathrm{cm}$. 5: the rotation applied to each bin; the peak gets $-78.8°$, the conjugate of the residual phase.

The rule that the map implements. A Doppler bin must be corrected with the residual of the scatterer whose energy it holds, and with nothing else. Where two scatterers are seen in the same window, the spectrum has two peaks, each at its own scatterer's cell, and each reads its own block of the map (Section 4, Fig. 9). Where no scatterer's energy can land, the map is zero.

3 Stage B: the Sub-Aperture Kernel

Figure 4 shows the loop with the line of code that executes each box. The subsections follow the boxes.

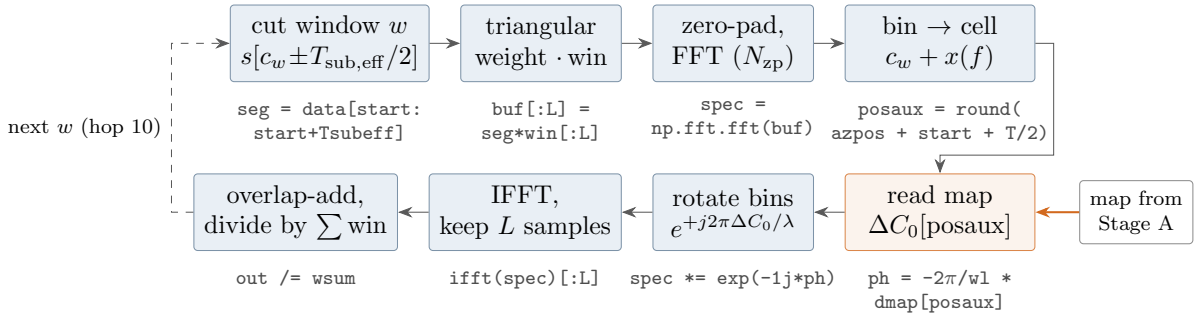


Figure 4: The kernel `sata_1d` for one channel line. Grey text: the statement that executes each box. The orange path is the only place where the map enters.

3.1 Sub-aperture length, hop and FFT size

The sub-aperture must be short enough for the residual to be constant over the ground patch it sees, and long enough for its Doppler resolution to separate positions. The classical topography-accommodation length is (report, Eq. 44)

$$\delta x = \sqrt{\frac{\lambda\, r_{\mathrm{ref}}}{2}}, \qquad T_{\mathrm{sub,eff}} = 2\left\lfloor \frac{\delta x\, \mathrm{PRF}_{\mathrm{ch}}}{2v} \right\rceil, \qquad \mathrm{hop} = \frac{T_{\mathrm{sub,eff}}}{2}, \qquad N_{\mathrm{zp}} = 2^{\lceil \log_2 T_{\mathrm{sub,eff}} \rceil} \cdot \mathrm{osf}. \tag{3}$$

The rate is that of the line actually processed, $\mathrm{PRF}_{\mathrm{ch}}$ (report, Section 4.2, point 1). Numerically, $\delta x = \sqrt{0.25 \cdot 766\,200/2} = 309.5\,\mathrm{m}$, which is $\delta x\, \mathrm{PRF}_{\mathrm{ch}}/v = 20.1$ samples, so $T_{\mathrm{sub,eff}} = 20$, hop $= 10$ and, with osf $= 4$, $N_{\mathrm{zp}} = 128$. The bin spacing is $\mathrm{PRF}_{\mathrm{ch}}/N_{\mathrm{zp}} = 3.906\,\mathrm{Hz}$. The oversampling only refines the bin grid; the result does not depend on it (test T16).

sata1d/sata.py, lines 25--30

```
25 def subaperture_sizing(rref, prf, v, wl, sata_osf=1):
26     """Sub-aperture length Tsubeff (even), hop Tsub (50 % overlap), FFT size Nzp."""
27     deltax = np.sqrt(wl * rref / 2.0)                    # SATA resolution on ground [m]
28     Tsubeff = int(np.round(deltax * prf / v * 0.5) * 2)
29     Nzp = int(2 ** np.ceil(np.log2(Tsubeff)) * sata_osf)
30     return Tsubeff, Tsubeff // 2, Nzp
```

3.2 Window and constant overlap-add

The analysis window is triangular, $\mathrm{win}[n] = n/(T_{\mathrm{sub,eff}}/2-1)$ on the rising half and mirrored on the falling half. At 50 % overlap the shifted windows sum to a constant (constant overlap-add), and the loop divides the accumulated output by that sum:

$$\tilde{s}[n] = \frac{\sum_w \mathrm{IFFT}\{S_w \cdot R_w\}[n - w\,\mathrm{hop}]}{\sum_w \mathrm{win}[n - w\,\mathrm{hop}]}, \qquad S_w = \mathrm{FFT}_{N_{\mathrm{zp}}}\{s \cdot \mathrm{win}\}, \tag{4}$$

where R_w is the per-bin rotation of Section 3.5. With $R_w \equiv 1$ the IFFT returns the windowed segment and Eq. (4) reduces to $\tilde{s} = s$: a null map reproduces the input to machine precision (T1).

sata1d/sata.py, lines 46--48

```
46 def triangular_window(Tsubeff):
47     half = np.arange(Tsubeff // 2) / (Tsubeff // 2 - 1)
48     return np.concatenate((half, half[::-1]))
```

3.3 The ruler: from a Doppler bin to an azimuth cell

This is the step that makes the correction position-dependent. Take a scatterer at cell x_t and a window centred at cell c_w (Fig. 5). The along-track distance between the phase centre and the scatterer is $d = (x_t - c_w)\, d_s$ and the range is $R = \sqrt{r^2 + d^2}$. The two-way Doppler is

$$f = -\frac{2}{\lambda}\frac{dR}{dt} = \frac{2v}{\lambda}\frac{d}{R} = \frac{2v}{\lambda}\sin\beta, \qquad \tan\beta = \frac{d}{r}, \tag{5}$$

positive while the platform approaches, i.e. while the scatterer is ahead of the window centre. The kernel inverts Eq. (5): a bin with frequency f belongs to the angle $\beta(f)$, hence to the distance $d = r\tan\beta(f)$, hence to the cell

$$\beta(f) = \arcsin\left(\frac{\lambda f}{2v}\right), \qquad x(f) = r\left[\tan\beta(f) - \tan\beta_{\text{img}}\right]\frac{\text{PRF}_{\text{ch}}}{v}, \qquad \text{posaux}(f) = c_w + x(f), \tag{6}$$

which is Eq. 45 of the report, with $\beta_{\text{img}} = 0$ for the broadside grid used here. The decisive property is that the ruler is the exact inverse of the physics that placed the energy: wherever the window is, the bin that holds the scatterer's energy satisfies $c_w + x(f) = x_t$. *The peak always points at the scatterer's own cell*, even though it moves through the spectrum from window to window (T6, Fig. 13, right).

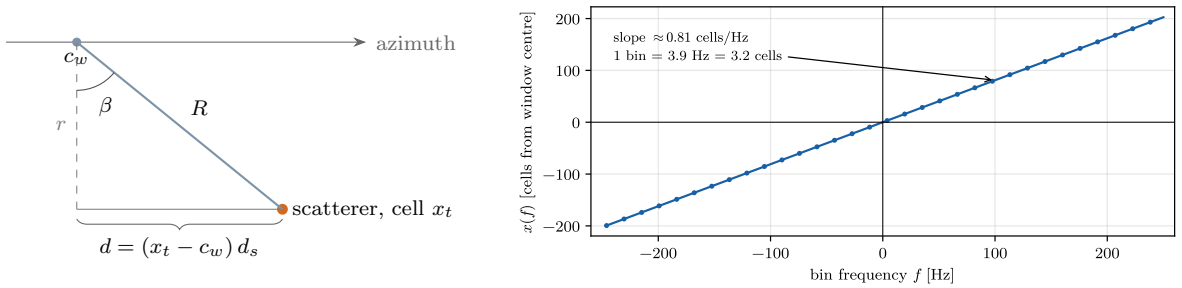


Figure 5: Left: geometry of Eq. (5); the scatterer seen from the window centre under the squint β. Right: the ruler $x(f)$ of Eq. (6) for the 128 bins of the kernel. It is almost linear, $x(f) \simeq (r\lambda\text{PRF}_{\text{ch}}/2v^2)\, f = 0.81$ cells/Hz, so the band $\pm\text{PRF}_{\text{ch}}/2$ spans ± 203 cells around the window centre and one bin spans 3.2 cells.

In code, `subaperture_axis` builds the bin frequencies in FFT order (equal to `np.fft.fftfreq` on all bins but the Nyquist one, which it labels $+\text{PRF}_{\text{ch}}/2$; T19) and their cell offsets:

sata1d/sata.py, lines 33--43

```
33 def subaperture_axis(prf, Nzp, v, wl, r, squint=0.0):
34     """Doppler of each FFT bin (np.fft.fft order) and the cell offset it points to:
35     f -> beta = asin(wl f / 2v) -> r (tan beta - tan squint) / v * prf."""
36     fc = 2.0 * v / wl * np.sin(squint)
37     pfc = np.round(np.mod(fc, prf) * Nzp / prf)
38     dfc = np.round(fc * Nzp / prf) * prf / Nzp
39     fsub = np.arange(Nzp) * prf / Nzp - prf * 0.5 + dfc
40     fsub[0] = prf * 0.5 + dfc
```

```
    fsub = np.roll(fsub, int(Nzp / 2 + pfc))
    betasub = np.arcsin(np.clip(wl * fsub / (2.0 * v), -1.0, 1.0))
    return fsub, r * (np.tan(betasub) - np.tan(squint)) / v * prf
```

3.4 Folded Doppler: the target also appears $\pm X$ cells away

A channel is sampled at $\mathrm{PRF}_{\mathrm{ch}} = 500\,\mathrm{Hz}$, but a target sweeps a Doppler band $B = 4v\sin(\theta_{az}/2)/\lambda = 1281\,\mathrm{Hz}$ (θ_{az}: azimuth beamwidth). A true Doppler f outside $[-\mathrm{PRF}_{\mathrm{ch}}/2, \mathrm{PRF}_{\mathrm{ch}}/2)$ is stored in the bin labelled $f + m\,\mathrm{PRF}_{\mathrm{ch}}$ for the integer m that brings it into the window. Since the ruler is practically linear, that label points

$$\begin{aligned} \mathrm{posaux} &= c_w + x(f + m\mathrm{PRF}_{\mathrm{ch}}) \simeq x_t + m\,X, \\ X &= x(\mathrm{PRF}_{\mathrm{ch}}) = r\tan\Big(\arcsin\frac{\lambda\mathrm{PRF}_{\mathrm{ch}}}{2v}\Big)\frac{\mathrm{PRF}_{\mathrm{ch}}}{v} = 405.06\ \text{cells} = 6.23\ \text{km}, \end{aligned} \tag{7}$$

that is, m folds away from the scatterer (the nonlinearity of tan contributes less than 0.02 cells). With $|f| \le B/2 = 641\,\mathrm{Hz}$ the whole-band kernel sees $m \in \{-1, 0, +1\}$: at the beginning of the illumination ($f > \mathrm{PRF}_{\mathrm{ch}}/2$) the peak points to $x_t - X$, in the middle to x_t, at the end to $x_t + X$ (Fig. 6, T6). The map must therefore also hold the scatterer's residual at $x_t \pm X$, which Stage A does with the alias layers of Section 4.4. The same fact has a consequence for scenes with several scatterers, treated in Section 6.3: two scatterers X apart share the same bins.

3.5 The rotation

Each bin reads the map at its cell and is multiplied by

$$R_w(f) = e^{-j\,\mathrm{ph}(f)}, \qquad \mathrm{ph}(f) = -\frac{2\pi}{\lambda}\,\Delta C_0[\mathrm{posaux}(f)] \quad\Longrightarrow\quad R_w(f) = e^{+j2\pi\Delta C_0[\mathrm{posaux}(f)]/\lambda}, \tag{8}$$

the conjugate of the unwanted factor in Eq. (1) (`inverse=True` removes, `inverse=False` injects). If the map is constant, every bin of every window gets the same factor and Eq. (4) returns $s\,e^{+j2\pi\Delta C_0/\lambda}$ exactly (T3). The sign is verified directly in T12: only this sign brings the reconstruction back to the reference; the opposite one doubles the error. The complete kernel is

sata1d/sata.py, lines 51--85

```
def sata_1d(data, delta_C0_array, rref, prf, v, wl, r, squint=0.0, inverse=False,
    sata_osf=1):
    """
    data           : (naz,) complex azimuth line of one channel at prf (= PRF_op)
    delta_C0_array : (naz,) residual range [m] per azimuth cell
    inverse=True   : remove the residual (spec *= exp(-j ph)); False injects it.
    """
    data = np.asarray(data, dtype=complex)
    dimx = len(data)
    Tsubeff, hop, Nzp = subaperture_sizing(rref, prf, v, wl, sata_osf)
    if Tsubeff <= 2:
        return data.copy()
    win = triangular_window(Tsubeff)

    fsub, azpos = subaperture_axis(prf, Nzp, v, wl, r, squint)

    out = np.zeros(dimx, dtype=complex)
    wsum = np.zeros(dimx)
    for start in range(0, dimx, hop):
```

```
        seg = data[start:start + Tsubeff]
        L = len(seg)
        if L < 2:
            break
        buf = np.zeros(Nzp, dtype=complex)
        buf[:L] = seg * win[:L]
        spec = np.fft.fft(buf)
        posaux = np.clip(np.round(azpos + start + 0.5 * Tsubeff).astype(int), 0, dimx - 1)
        ph = -2.0 * np.pi / wl * delta_C0_array[posaux]
        ph[~np.isfinite(ph)] = 0.0
        spec *= np.exp(-1j * ph) if inverse else np.exp(1j * ph)
        out[start:start + L] += np.fft.ifft(spec)[:L]
        wsum[start:start + L] += win[:L]
    nz = wsum > 1e-12
    out[nz] /= wsum[nz]
    out[~nz] = data[~nz]
    return out
```

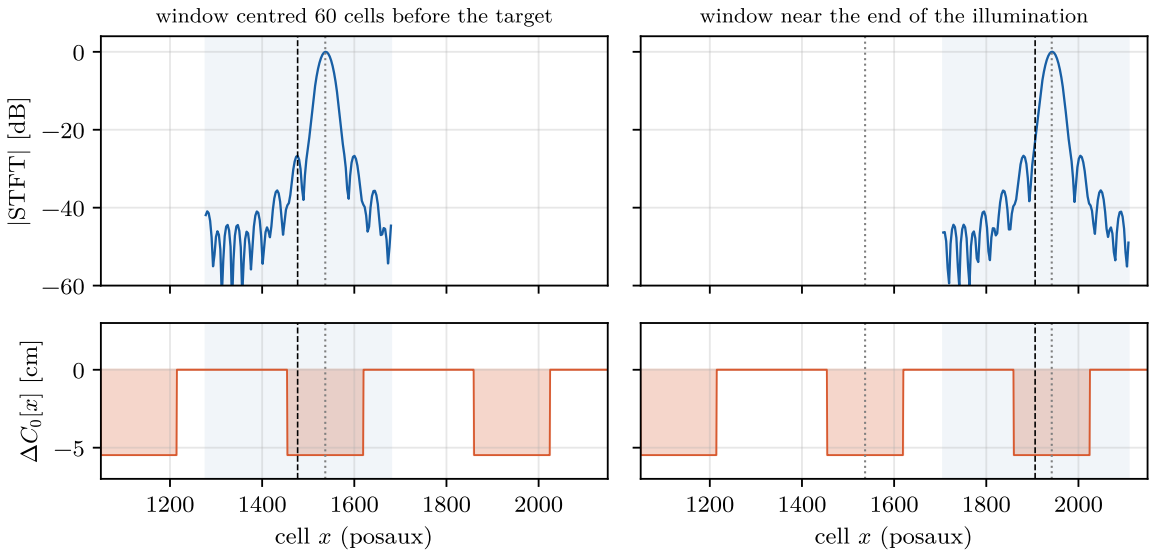


Figure 6: Two windows of channel 0 for the target at $h = 240\,\mathrm{m}$ (cell 1537). Top: |STFT| of the window, every bin drawn at its cell $c_w + x(f)$; the shaded band is the ± 203 cells the window can reach; dashed: window centre; dotted: x_t and $x_t + X$. Bottom: the map $\Delta C_0[x]$ of Stage A. Left: the window is 60 cells before the target, the peak sits at x_t and reads $-5.47\,\mathrm{cm}$. Right: near the end of the illumination the true Doppler exceeds $+\mathrm{PRF}_{\mathrm{ch}}/2$, the peak is folded to $x_t + X = 1942$ and reads the alias block there, which carries the same value.

4 Stage A: the ΔC_0 Map

The kernel reads ΔC_0[posaux] for every bin, including bins that hold no energy. A bin's rotation matters only in proportion to its energy, so the map has to be right where energy appears and can be anything elsewhere. The implementation writes each scatterer's residual on the cells where its energy can land and zero elsewhere. Figure 7 shows the five steps.

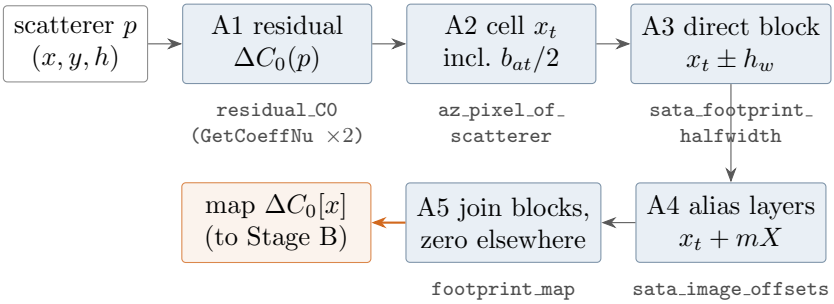


Figure 7: Stage A, `build_delta_C0_array`, once per channel (and, per sub-band, once per channel and band).

4.1 A1 — the residual of each scatterer

The residual is evaluated with the same semi-numerical model that builds the filters, so that error and correction are consistent by construction (report, Eq. 36). `GetCoeffNu` fits two parabolas over the common illuminated aperture: the monostatic history $R_{ms}(t) = 2R_{Tx}(t) \simeq c_0 + c_1 t + c_2 t^2$ and the bistatic-minus-monostatic difference $R_{bs}(t) - R_{ms}(t) \simeq \Delta c_0 + \Delta c_1 t + \Delta c_2 t^2$. The value of a parabola at its vertex is $c_0 - c_1^2/4c_2$, and C_0 is the difference of the two vertex values:

$$C_0 = \left[(c_0 + \Delta c_0) - \frac{(c_1 + \Delta c_1)^2}{4(c_2 + \Delta c_2)}\right] - \left[c_0 - \frac{c_1^2}{4c_2}\right] = \Delta c_0 + \frac{c_1^2}{4c_2} - \frac{(c_1 + \Delta c_1)^2}{4(c_2 + \Delta c_2)}. \tag{9}$$

`residual_C0` calls it twice, for the scatterer and for the reference point, and returns the difference:

sata1d/sata.py, lines 91--96

```
91 def residual_C0(cfg: ExperimentConfig, tracks: PlatformTracks, ptg_real, channel):
92     """C0(scatterer) - C0(reference point) for one channel [m]."""
93     common = (tracks.ptx, tracks.prx[channel], tracks.vtx, tracks.vrx[channel],
94               tracks.ptx, tracks.vtx, cfg.prf, cfg.system.wl, cfg.ta,
95               cfg.sq_tx, cfg.sq_rx[channel], cfg.theta_tx, cfg.theta_rx[channel])
96     return float(GetCoeffNu(ptg_real, *common)[0] - GetCoeffNu(cfg.scene.ptg, *common)[0])
```

For the target at $h = 240\,\mathrm{m}$ this gives $\Delta C_0 = [-5.474,\ -2.319,\ -0.352,\ -0.142]\,\mathrm{cm}$ for the four channels, within 0.3 % of Eq. (2) (T4, Fig. 13).

4.2 A2 — where the scatterer sits on the channel line

The channel line is indexed by azimuth cells of $d_s = v/\mathrm{PRF}_{\mathrm{ch}}$, with the scene centre at $N_{a,ch}/2$. The receiver of channel i trails the transmitter by $b_{at,i}$ (`geometry.py`), so the Tx–Rx phase centre trails it by $b_{at,i}/2$ and the closest approach to scatterer $x_0 + \Delta x$ falls at

$$x_t = \left\lfloor \frac{N_{a,ch}}{2} + \frac{x_0 + \Delta x + b_{at,i}/2}{d_s} \right\rceil. \tag{10}$$

For the DPCA array ($b_{at}/2 = 0,\ 3.84,\ 7.69,\ 11.53\,\mathrm{m}$) this is cell 1537 for channel 0 and 1538 for the others, identical to the minimum of the simulated bistatic range history (T5).

sata1d/sata.py, lines 99--103

```
99 def az_pixel_of_scatterer(cfg: ExperimentConfig, dx, channel):
100     """Cell of scatterer x0 + dx on the per-channel line; the Tx-Rx phase centre
101     trails the transmitter by bat/2."""
102     ds = cfg.system.vs / cfg.PRF_op
103     return int(round(cfg.Na_ch / 2 + (cfg.scene.x0 + dx + 0.5 * cfg.array.bat[channel]) /
        ds))
```

4.3 A3 — the footprint: how far a scatterer's energy spreads

Within one window the scatterer is a nearly pure tone: its Doppler changes by $K_a\, T_{\mathrm{sub,eff}}/\mathrm{PRF}_{\mathrm{ch}} = 617\,\mathrm{Hz/s} \cdot 0.04\,\mathrm{s} \approx 25\,\mathrm{Hz}$. Its spectrum is therefore the transform of the triangular window, $|W(f)| \propto \mathrm{sinc}^2(f\, T_{\mathrm{sub,eff}}/2\mathrm{PRF}_{\mathrm{ch}})$, centred on its instantaneous Doppler, with nulls at multiples of $2\mathrm{PRF}_{\mathrm{ch}}/T_{\mathrm{sub,eff}} = 50\,\mathrm{Hz}$. The map covers the response up to the second null,

$$f_{\mathrm{edge}} = \frac{4\,\mathrm{PRF}_{\mathrm{ch}}}{T_{\mathrm{sub,eff}}} = 100\ \mathrm{Hz}, \qquad h_w = \lceil x(f_{\mathrm{edge}}) \rceil = 82\ \text{cells}\ (1.26\ \mathrm{km}), \tag{11}$$

and every bin that holds energy of the scatterer then points inside $x_t \pm h_w$. T15 compares the nulls: the first (41 cells) leaves 3.0 % of error on the corrected channel, the second 2.0 %, the third 1.7 %; all give the same ambiguity after reconstruction. The second null covers 99.9 % of the energy (T8) and keeps the blocks narrow, which matters for the collisions of Section 6.3.

sata1d/sata.py, lines 106--115

```
106 def sata_footprint_halfwidth(cfg: ExperimentConfig, squint=0.0):
107     """Half-width [cells] of one scatterer in a sub-aperture spectrum: triangular
108     window up to its 2nd null (4 PRF_op / Tsubeff), mapped through the ruler."""
109     prf, v, wl, r = cfg.PRF_op, cfg.system.vs, cfg.system.wl, cfg.scene.r0
110     Tsubeff, _, _ = subaperture_sizing(r, prf, v, wl)
111     f_edge = 4.0 * prf / Tsubeff
112     fc = 2 * v / wl * np.sin(squint)
113     b0 = np.arcsin(np.clip(wl * fc / (2 * v), -1, 1))
114     b1 = np.arcsin(np.clip(wl * (fc + f_edge) / (2 * v), -1, 1))
115     return int(np.ceil(r * (np.tan(b1) - np.tan(b0)) / v * prf))
```

4.4 A4 — alias layers

From Section 3.4, the same energy also lands at $x_t + mX$. For a kernel whose labels cover $[f_c - \mathrm{PRF}_{\mathrm{ch}}/2,\ f_c + \mathrm{PRF}_{\mathrm{ch}}/2)$ the folds that occur are

$$m \in \left\{ \left\lfloor \frac{f_c - B/2}{\mathrm{PRF}_{\mathrm{ch}}} \right\rceil, \ \ldots, \ \left\lfloor \frac{f_c + B/2}{\mathrm{PRF}_{\mathrm{ch}}} \right\rceil \right\}, \tag{12}$$

which gives $\{-1, 0, 1\}$ for the whole band ($f_c = 0$) and one-sided sets for the sub-bands ($f_c = f_k$, Section 5). The alias layers are not a convenience: the reconstruction cancels the folded components of a scatterer in the other output bands only if those components carry channel phases consistent with the model, i.e. only if they are corrected with the scatterer's own residual too.

sata1d/sata.py, lines 118--129

```
118 def sata_image_offsets(cfg: ExperimentConfig, f_centre=0.0):
119     """Offsets [cells] where one scatterer appears for a kernel whose bins are labelled
120     with the absolute Doppler in [f_centre - PRF_op/2, f_centre + PRF_op/2).
121     A true Doppler f (|f| <= B/2, B = 4 v sin(theta/2)/wl) is labelled f + m PRF_op,
122     so it lands m*X cells away, X = cell shift of one PRF_op fold.
123     Whole band (f_centre = 0): {-X, 0, +X}; sub-band k: one-sided set."""
124     prf, v, wl, r = cfg.PRF_op, cfg.system.vs, cfg.system.wl, cfg.scene.r0
125     X = r * np.tan(np.arcsin(np.clip(wl * prf / (2 * v), -1, 1))) / v * prf
126     B = 4.0 * v * np.sin(cfg.theta_tx / 2.0) / wl
127     m_lo = int(np.round((f_centre - B / 2) / prf))
128     m_hi = int(np.round((f_centre + B / 2) / prf))
129     return np.array([m * X for m in range(m_lo, m_hi + 1)], dtype=float)
```

4.5 A5 — drawing the map

`footprint_map` draws the map in three steps (Fig. 8):

1. *Terrain blocks.* Scatterers whose footprints touch (gap $\leq 2h_w$) form one block, linearly interpolated between their values over $[x_{\text{first}} - h_w,\ x_{\text{last}} + h_w]$; an isolated scatterer gives a constant block $x_t \pm h_w$.
2. *Images.* Every block is drawn at each offset of Eq. (12): the direct image ($m = 0$) and the folded copies (mX). Where two images overlap, a cell takes the image whose core $[x_{\text{first}}, x_{\text{last}}] + mX$ is *nearest* (ties go to the direct image, then to the smaller $|m|$). Each image is the footprint of a component whose main lobe is centred on its core, so the nearest-image rule gives every main lobe its own value and splits the overlap at the midpoint between the two components. This rule replaces an earlier one, "the direct image wins the whole overlap", which let the block of one scatterer overwrite the folded main lobe of another up to h_w away and degraded scenes with separations near $X \pm h_w$ (Section 6.3, T17b).
3. *Zero* on every other cell. Those bins hold no scatterer energy.

sata1d/sata.py, lines 132--163

```python
def footprint_map(values_by_pixel: dict, naz: int, halfwidth: int, offsets=(0.0,)):
    """
    1) terrain blocks: scatterers whose footprints touch (gap <= 2 hw) form one block,
       linearly interpolated between them over [first - hw, last + hw]; an isolated
       scatterer gives a constant block pixel +- hw;
    2) images: every block is drawn at each offset (0 = direct, m X = folded copies);
       where images overlap, a cell takes the image whose core [first, last] + offset
       is nearest (ties: the direct image, then the smaller |offset|), so that the
       main lobe of each component reads its own value;
    3) zero elsewhere.
    """
    arr = np.zeros(naz)
    if not values_by_pixel:
        return arr
    pix = np.array(sorted(values_by_pixel), dtype=float)
    val = np.array([max(values_by_pixel[p], key=abs) for p in sorted(values_by_pixel)])
    cuts = np.flatnonzero(np.diff(pix) > 2 * halfwidth) + 1
    blocks = list(zip(np.split(pix, cuts), np.split(val, cuts)))
    dist = np.full(naz, np.inf)
    for off in sorted(offsets, key=abs):
        sh = float(np.round(off))
        for bp, bv in blocks:
            lo = int(max(0, np.floor(bp[0] + sh - halfwidth)))
            hi = int(min(naz - 1, np.ceil(bp[-1] + sh + halfwidth)))
            if lo > hi:
                continue
            x = np.arange(lo, hi + 1)
            d = np.maximum(0.0, np.maximum(bp[0] + sh - x, x - (bp[-1] + sh)))
            take = d < dist[x]
            arr[x[take]] = np.interp(x[take] - sh, bp, bv)
            dist[x[take]] = d[take]
    return arr
```

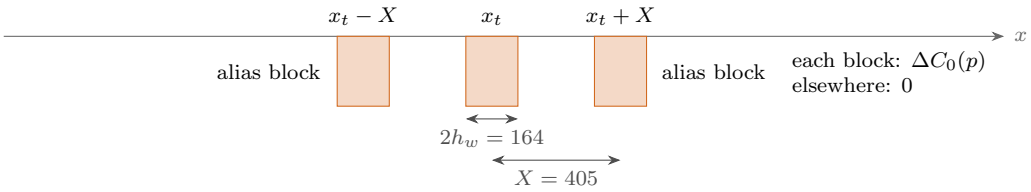


Figure 8: Map of one isolated scatterer for the whole-band kernel (channel 0: $x_t = 1537$, value -5.47 cm). The direct block receives the peak in the central part of the illumination, the alias blocks the peaks at the two ends (Fig. 6).

Two scatterers with overlapping apertures. Where the illuminated spans of two scatterers overlap, a window sees both. Its spectrum then has two peaks, one per scatterer, and by Section 3.3 each peak points at its own scatterer's cell. Each therefore reads its own block of the map and is rotated with its own residual (Fig. 9). T13 measures this directly: with the map built from both targets, the signal of each target alone is corrected to within 2.2 % of its own ideal rotation. The previous map, which held the value of the nearest scatterer over the whole line (`mode="hold"`), gave the right phase to one of the two peaks only.

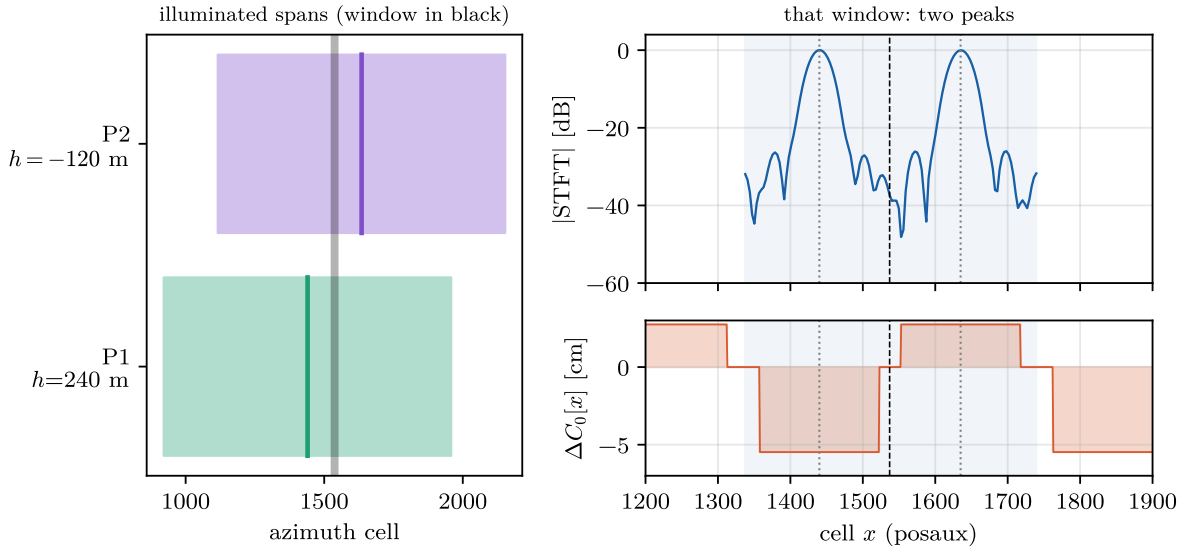


Figure 9: P1 ($h = 240$ m, $\Delta C_0 = -5.47$ cm) and P2 ($h = -120$ m, $\Delta C_0 = +2.74$ cm), 3 km apart, channel 0. Left: their illuminated spans overlap; the window marked in black sees both. Right: the spectrum of that window has two peaks, one at each scatterer's cell (dotted), and each falls in its own block of the map. The blocks at the edges are the alias layers of the other scatterer ($x_2 - X$, $x_1 + X$).

Two limits set by the sub-aperture. The spectrum of a window separates two scatterers only if their peaks are more than about one resolution cell apart, that is, if the scatterers are more than $\delta x \simeq 309$ m apart (20 cells). Closer scatterers share bins, and the joined block interpolates between their residuals: exact for a smooth slope, approximate for a jump. And two scatterers whose separation is close to a multiple of X share bins for a different reason, the fold of Section 3.4; that case is analysed in Section 6.3. Both limits belong to the method, not to the implementation.

5 Whole Band and per Sub-Band

5.1 What changes

The reconstruction delivers N_{rx} Doppler sub-bands; sub-band k is centred at (report, Eq. 46)

$$f_k = \Big(-\frac{N_{rx}}{2} + k + \frac{1}{2}\Big)\mathrm{PRF}_{\mathrm{ch}} = -750,\ -250,\ +250,\ +750\ \mathrm{Hz}, \qquad \beta_k = \arcsin\frac{\lambda f_k}{2v}, \tag{13}$$

and `reconstruct_from_spectra` indexes its output bands in the same order (band k covers $[-\mathrm{PRF}/2 + k\mathrm{PRF}_{\mathrm{ch}},\ -\mathrm{PRF}/2 + (k+1)\mathrm{PRF}_{\mathrm{ch}})$), which was checked against `subband_frequency_beam`. The *whole-band* variant corrects each channel once and feeds that single set to all sub-bands. The *per-sub-band* variant repeats both stages for every k (Fig. 10):

- A1 fits ΔC_0 only over the look angles of the band, $[\beta(f_k - \mathrm{PRF}_{\mathrm{ch}}/2),\ \beta(f_k + \mathrm{PRF}_{\mathrm{ch}}/2)]$ (`getcoeff_beam`);
- the kernel labels its bins with the absolute Doppler of band k (below);
- A4 uses the fold set of Eq. (12) with $f_c = f_k$;
- sub-band k of the reconstruction uses only the set corrected for k.

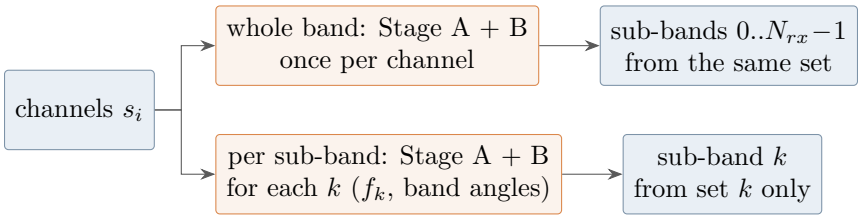


Figure 10: The two variants. The per-sub-band one costs N_{rx} corrections and N_{rx} reconstructions instead of one of each.

5.2 The sub-band axis

The kernel of band k must label every bin with the absolute Doppler it holds within $[f_k - \mathrm{PRF}_{\mathrm{ch}}/2,\ f_k + \mathrm{PRF}_{\mathrm{ch}}/2)$ and measure positions on the broadside grid on which the map is defined (report, Section 4.2, points 2 and 3). A baseband bin ϕ (`fftfreq`) holds the true Dopplers $\phi + m\mathrm{PRF}_{\mathrm{ch}}$, and exactly one of them lies in the window of band k:

$$\begin{aligned} f_{\mathrm{lo}} &= f_k - \frac{\mathrm{PRF}_{\mathrm{ch}}}{2}, \qquad \tilde{f}(\phi) = f_{\mathrm{lo}} + \big[(\phi - f_{\mathrm{lo}}) \bmod \mathrm{PRF}_{\mathrm{ch}}\big], \\ x(\tilde{f}) &= r\tan\beta(\tilde{f})\,\frac{\mathrm{PRF}_{\mathrm{ch}}}{v} \qquad (\beta_{\mathrm{img}} = 0). \end{aligned} \tag{14}$$

Energy of band k then points straight at x_t, while energy of the other bands points at $x_t + mX$ with m from Eq. (12). Figure 11 shows how the map moves its alias layers with k. T19 checks that every axis lies inside its band and is congruent to `fftfreq` modulo $\mathrm{PRF}_{\mathrm{ch}}$.

sata1d/subband.py, lines 94--101

```
94  def subband_axis(prf, Nzp, v, wl, r, f_k):
95      """Absolute Doppler of each FFT bin for sub-band k (baseband frequency wrapped into
96      [f_k - prf/2, f_k + prf/2)) and its cell offset on the broadside grid."""
97      phi = np.fft.fftfreq(Nzp, d=1.0 / prf)
98      f_lo = f_k - prf / 2.0
99      fsub = f_lo + np.mod(phi - f_lo, prf)
100     betasub = np.arcsin(np.clip(wl * fsub / (2.0 * v), -1.0, 1.0))
101     return fsub, r * np.tan(betasub) / v * prf
```

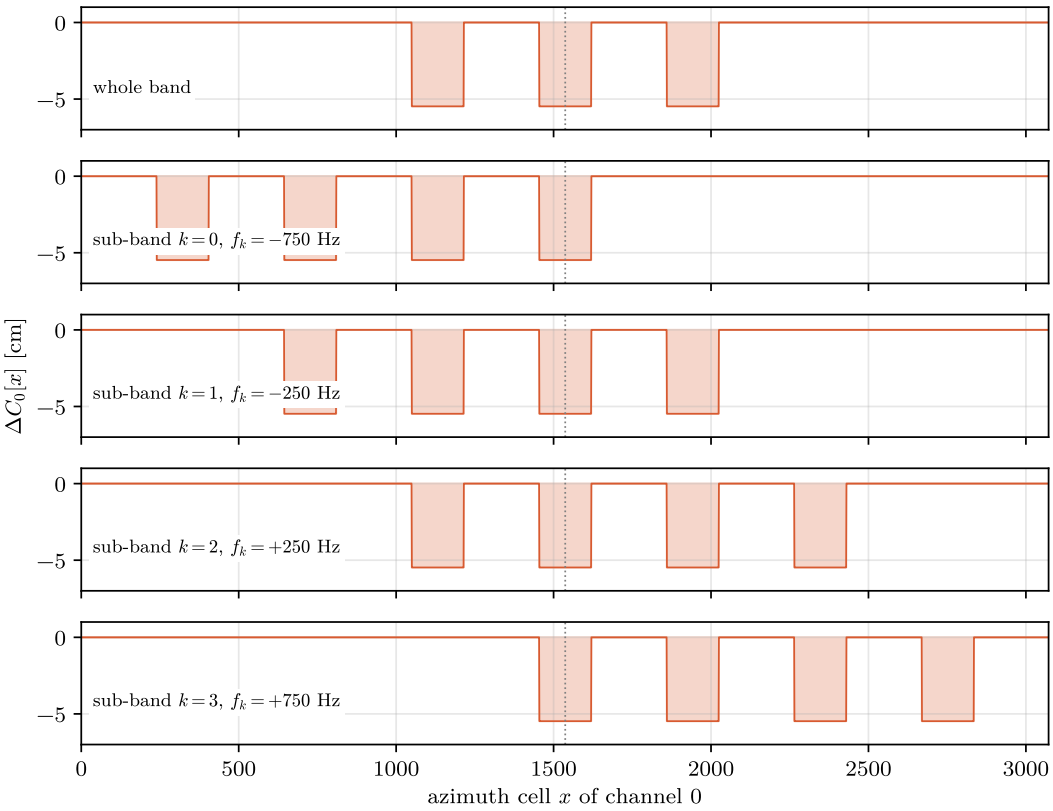


Figure 11: Maps of channel 0 for the whole band and the four sub-bands. The direct block at $x_t = 1537$ is always present; the alias layers follow the fold set of each band: $\{-1, 0, 1\}$ for the whole band and $\{-3..0\}$, $\{-2..1\}$, $\{-1..2\}$, $\{0..3\}$ for $k = 0..3$. The value is the same in all bands because ΔC_0 of an iso-range target does not depend on the look angle.

5.3 Correction of the previous per-sub-band axis

The previous one-dimensional per-sub-band kernel used $f = f_k + \phi$ and positions measured from the squint of the band, $x = r[\tan\beta(f_k+\phi) - \tan\beta_k]\,\mathrm{PRF}_{\mathrm{ch}}/v$. Both terms are inconsistent with Eq. (14). Since f_k is an odd multiple of $\mathrm{PRF}_{\mathrm{ch}}/2$, the label $f_k+\phi$ is not one of the Dopplers $\phi + m\mathrm{PRF}_{\mathrm{ch}}$ that bin ϕ actually holds; subtracting $\tan\beta_k$ then shifts the grid back by the same half band. To first order the two errors cancel, $x \simeq 0.81\,\phi$, and the "per-sub-band" kernel was position-for-position the whole-band kernel: the energy of band k pointed at x_t, $x_t \pm X$ depending on the fold (Fig. 12, left; T7: up to 407 cells, 60 % of the windows off by at least one fold), and read the right value only because its map had the whole-band alias set. This is the "axis of the right width arising from two compensating errors" described in the report. With Eq. (14) every window of band k points at the target to within one bin (right panel, max. 2.4 cells). For the iso-range scenes simulated here the two kernels give the same reconstruction (Table 1): within 0.5 dB on ordinary scenes and within 1.8 dB near an alias collision, with no systematic direction. This is what the algebra predicts, since `residual_C0_subband` equals `residual_C0` to 10^{-9} m for an iso-range target and every image of a target in the map then carries the same value, whichever fold a bin is attributed to. The correction is justified by the derivation and by T7, not by a gain on these scenes; the two kernels differ when the residual varies with the look angle within the aperture.

Table 1: Old and corrected sub-band axis, same maps: worst ambiguity after reconstruction [dB].

scene	ref.	no SATA	whole	sub, corrected axis	sub, previous axis
1 target, $h = 240$ m	−39.1	−13.1	−39.4	−39.4	−39.5
2 targets 3 km (240 / −120 m)	−37.9	−12.8	−38.3	−38.3	−38.8
ramp 9×200 m, 15°	−33.9	−8.6	−30.8	−30.8	−30.4
spike 5×200 m, 500 m	−34.3	−7.6	−13.1	−13.1	−13.1
2 targets 5.5 km (collision edge)	−37.9	−12.3	−30.2	−28.4	−30.8
2 targets 6.0 km (collision core)	−37.9	−12.2	−22.6	−22.3	−24.1

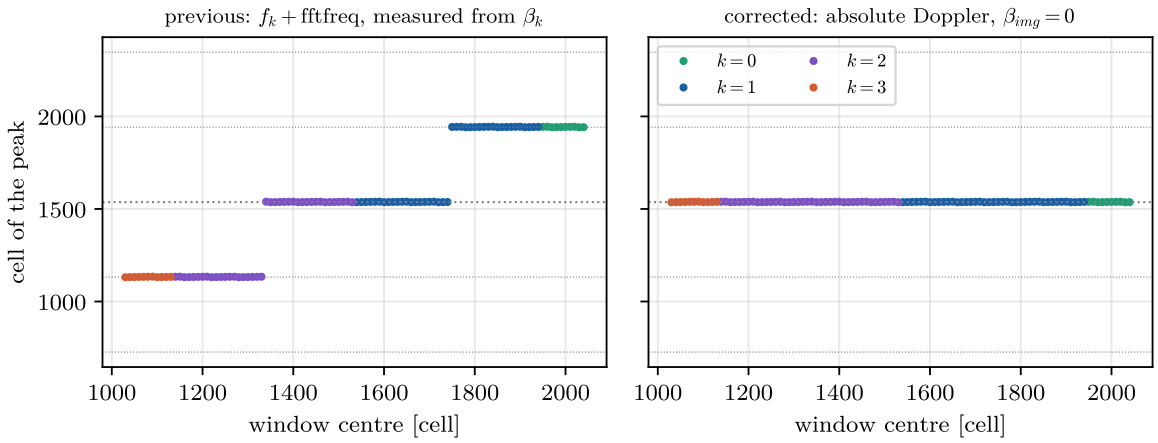


Figure 12: Cell pointed to by the STFT peak, for the windows whose instantaneous Doppler lies in band k (colours). Dotted: x_t and $x_t + mX$. Left: previous axis. Right: corrected axis of Eq. (14).

The kernel of band k is `sata_1d` with that axis, and the per-sub-band reconstruction assembles the N_{rx} corrected sets:

`sata1d/subband.py, lines 137--158`

```
137 def sata_channels_subband(cfg, tracks, s_channel, k, remove=True, sata_osf=4,
        mode="footprint"):
138     """Every channel SATA-corrected for output sub-band k."""
139     f_k = subband_frequency_beam(cfg, k)[0]
140     out = np.asarray(s_channel, dtype=complex).copy()
141     for kk in range(cfg.Nrx):
142         dC0 = build_delta_C0_subband_array(cfg, tracks, kk, k, mode=mode)
143         out[kk] = sata_1d_subband(out[kk], dC0, rref=cfg.scene.r0, prf=cfg.PRF_op,
        v=cfg.system.vs,
144                                   wl=cfg.system.wl, r=cfg.scene.r0, f_k=f_k,
145                                   inverse=remove, sata_osf=sata_osf)
146     return out
147
148
149 def reconstruct_subband(cfg: ExperimentConfig, tracks: PlatformTracks, s_channel,
        use_sata=True,
150                         remove=True, sata_osf=4, mode="footprint", zeroOutBw=True):
151     """Reconstruction where sub-band k uses the channels corrected for sub-band k.
152     use_sata=False gives the standard reconstruction."""
153     base = np.asarray(s_channel, dtype=complex)
154     spec = []
155     for k in range(cfg.Nrx):
156         corr = sata_channels_subband(cfg, tracks, base, k, remove, sata_osf, mode) if
        use_sata else base
157         spec.append([channel_spectrum(cfg, corr[j]) for j in range(cfg.Nrx)])
```

```
158     return reconstruct_from_spectra(cfg, tracks, spec, zeroOutBw)
```

6 Verification

The verification has three layers: a review of the conventions the correction depends on (Section 6.1), a suite of twenty tests on the reference configuration (Section 6.2, `verify_sata1d.py`: Table 3 of the report, $N_{rx} = 4$, DPCA along-track baselines, $b_{xt} \sim U(0, 100)$ m, seed 0), and the geometry sweep of Section 7. Every number below comes from one run of the suite; the figures are in `verification/`.

6.1 Conventions checked against the geometry and the reconstruction

Table 2: Sign and index conventions. Each was traced through the code and confirmed by a test.

Convention	Where it is fixed	Confirmed by
Rx_i trails Tx by $b_{at,i}$, so the phase centre trails by $b_{at,i}/2$	`geometry.py`: `prx[i,:,0] = vs*ta - bat[i]`	T5 (0 cells)
Data carry $e^{-j2\pi(R_{tx}+R_{rx})/\lambda}$; filters carry $e^{-j2\pi C_0/\lambda}$ of p_{ref}	`signal_model.py`, `filter_matrix`	T12 (sign)
Removal multiplies by $e^{+j2\pi\Delta C_0/\lambda}$	`sata_1d`, `inverse=True`	T3, T9, T12
Approaching $\Rightarrow f > 0 \Rightarrow$ scatterer ahead $\Rightarrow x(f) > 0$	Eq. (5), `subaperture_axis`	T6 (2.4 cells)
Kernel bins in FFT order, Nyquist labelled $+\mathrm{PRF}_{\mathrm{ch}}/2$	`subaperture_axis`	T19
Output band k of the reconstruction $= [-\mathrm{PRF}/2 + k\mathrm{PRF}_{\mathrm{ch}},\ \cdot + \mathrm{PRF}_{\mathrm{ch}})$	`reconstruct_from_spectra` (`fsub`) vs `subband_frequency_beam`	code trace, T10
Sub-band axis: absolute Doppler in band k, $\beta_{\mathrm{img}} = 0$	`subband_axis`	T7, T19
ΔC_0 from the same model as the filters (report Eq. 36)	`residual_C0` $\rightarrow$ `GetCoeffNu`	T4
$T_{\mathrm{sub,eff}}$ from δx at the rate of the processed line (Eq. 44)	`subaperture_sizing`($\mathrm{PRF}_{\mathrm{ch}}$)	$T_{\mathrm{sub,eff}} = 20$
Triangular window, 50 % overlap, COLA	`triangular_window`, `wsum`	T1

One cosmetic point was noted and left as is, to stay bit-identical with the reference code: in `getRawData1D` the extrapolation of the last sample of the receiver squint uses the transmitter array (`inst_sq_tx[Na-3]`); it affects one sample outside the illumination window.

6.2 The tests

Table 3: Verification tests (`python verify_sata1d.py`). Reference configuration; "5 kernels" means whole band plus the four sub-bands.

	What is checked	Criterion	Result	
T1	zero map returns the input (5 kernels)	rel. err. $< 10^{-12}$	1.2×10^{-16}	pass
T2	inject then remove the pipeline maps (5 kernels)	rel. err. $< 2\,\%$	$0.79\,\%$	pass
T3	constant map = one global rotation $e^{+j2\pi\Delta C_0/\lambda}$	rel. err. $< 10^{-12}$	1.4×10^{-16}	pass
T4	`residual_C0` vs closed form, 4 heights $\times$ 4 channels	ratio within $2\,\%$	0.9976–0.9998	pass
T5	cell of Eq. (10) vs simulated closest approach	≤ 1 cell	0 cells	pass
T6	whole band: STFT peak $\to x_t + mX$ (102 windows)	≤ 1 bin (3.2 cells)	2.4 cells	pass
T7	per sub-band: own-band peak $\to x_t$	≤ 1 bin	2.4 cells	pass
T8	fraction of the target's STFT energy read inside the map	$\geq 99\,\%$	99.91 % (all 5)	pass
T9	corrected channel vs $e^{+j2\pi\Delta C_0/\lambda}\,s$, 4 ch. $\times$ 5 kernels	$< 5\,\%$	$1.97\,\%$	pass
T10	ambiguity after reconstruction, h = 0..400 m	within 1 dB of ref.	0.27 dB	pass
T11	multi-target scenes vs oracle	reported	Table 4	—
T12	sign: remove / inject / remove twice	only "remove" reaches ref.	-39.4 / -7.8 / -13.1 dB	pass
T13	cross-talk: two-target map applied to each target alone	own rotation, $< 5\,\%$	$2.23\,\%$	pass
T14	channel-level error vs oracle channels (T11 scenes)	SATA < previous map	all 7 scenes	pass
T15	footprint half-width: 1st–4th null	reported	3.0 / 2.0 / 1.7 / 1.5 %	—
T16	kernel oversampling osf $= 1, 2, 4, 8$	within 1 dB	0.00 dB	pass
T17a	random sparse scenes, 3 targets 3.8–4.9 km apart, vs oracle	within 1 dB	0.33 dB	pass
T17b	two targets vs separation d, outside $\lvert d - mX\rvert \leq h_w$	within 1 dB of oracle	0.24 dB	pass
T18	$b_{xt} = 0$: map is zero and SATA is the identity	$< 10^{-8}$	3×10^{-12} m, 5×10^{-11}	pass
T19	kernel axis = fftfreq but Nyquist; sub-band axes in band	exact	127/128 bins, all 4 bands	pass

Kernel algebra (T1–T3, T16). T1 and T3 hold to machine precision, as Eq. (4) predicts. T2 holds to 0.8 % for the maps the pipeline produces: injection and removal are not exact inverses, because the windows overlap and the re-analysed spectra mix neighbouring bins. A white-noise map (±30 cm per cell) comes back with 96 % error; the correction can only represent a residual that varies slowly on the scale of δx, which is what SATA is designed for. The oversampling of the FFT (T16) changes the result by less than 0.01 dB from osf = 1 to 8.

Residual, position, ruler (T4–T7). Figure 13 compares the numerical residual with Eq. (2); Fig. 13 (right) tracks the STFT peak of every window across the illumination: $x_t - X$ at the start, x_t in the middle, $x_t + X$ at the end, with at most 2.4 cells of scatter, below the 3.2-cell width of one bin.

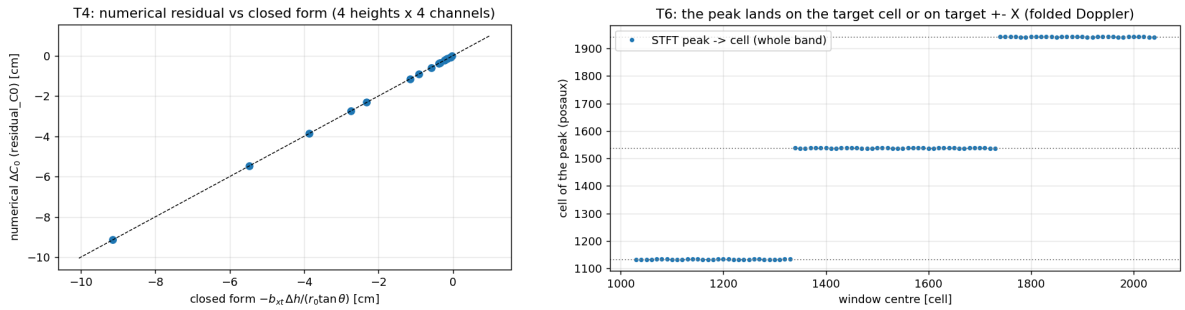


Figure 13: Left, T4: numerical ΔC_0 against the closed form, four heights (40–400 m) and four channels. Right, T6: cell of the STFT peak against the window centre, channel 0, whole-band kernel.

The sign (T12). The same map is applied with the removal sign, with the injection sign, and twice with the removal sign. Only the removal brings the ambiguity to the reference (-39.4 against -39.1 dB); injection doubles the residual and gives -7.8 dB, worse than no correction (-13.1 dB); removing twice leaves a residual of the same magnitude and opposite sign, and returns exactly the uncorrected level (-13.1 dB). The convention of Eq. (8) is therefore the only consistent one.

Correction and reconstruction (T8–T10). Of the target's STFT energy, 99.91 % falls on non-zero cells of the map, and the corrected channel is within 2 % of the ideal rotation; the remaining 2 % is energy in the sidelobes beyond $\pm h_w$, which reads zero. After reconstruction the worst ambiguity is -39.4 dB at all heights, against -39.1 dB for the reference (the scene reconstructed with the true height in the filters) and -22.9 to -9.0 dB without correction (Fig. 14). Whole band and per sub-band coincide, as expected for a single iso-range target.

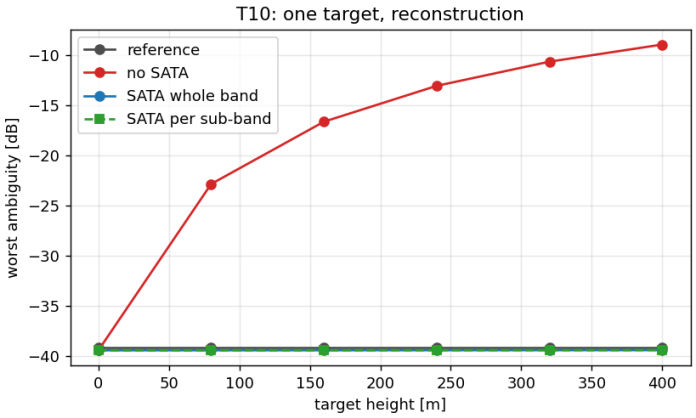


Figure 14: T10: worst azimuth ambiguity against target height, one target.

Several targets (T11, T13, T14). The benchmark for a scene with several scatterers is the *oracle*: each scatterer's signal is corrected with its own ΔC_0 before the channels are summed, which is what a perfect position-selective correction would achieve. Table 4 compares it with SATA, with the previous map and with no correction, after reconstruction; T14 does the same on the channels themselves, before reconstruction.

Table 4: T11 (worst azimuth ambiguity after reconstruction [dB], lower is better) and T14 (relative error of the corrected channels against the oracle channels).

	T11: ambiguity [dB]						T14: channel error [%]		
scene	ref.	oracle	whole	sub	prev. map	no SATA	SATA	prev.	none
2 targets, 3 km (240 / −120 m)	−37.9	−38.3	−38.3	−38.3	−15.1	−12.8	1.4	52.8	55.9
3 targets, 3 km (0 / 300 / 0 m)	−38.4	−38.5	−38.7	−38.6	−16.8	−11.2	1.3	37.7	48.5
ramp 5 × 200 m, 3°	−34.3	−31.0	−31.0	−31.0	−31.0	−28.6	0.6	6.3	8.0
ramp 9 × 200 m, 15°	−33.9	−30.8	−30.8	−30.8	−19.3	−8.6	3.2	53.7	68.6
random 5 × 200 m (0–100 m)	−34.3	−31.0	−31.1	−31.1	−26.6	−21.3	4.3	10.8	16.2
cliff 5 × 200 m, 100 m jump	−34.3	−31.0	−31.1	−31.1	−25.7	−20.8	5.1	17.3	23.8
spike 5 × 200 m, 500 m	−34.3	−31.0	−13.1	−13.1	−10.2	−7.6	36.5	45.9	51.1

Three observations follow. First, SATA reaches the oracle to within 0.3 dB in six of the seven scenes, and its channels are within 0.6–5 % of the oracle channels; the previous map loses more than 20 dB on the sparse scenes, because it gives the residual of one scatterer to the other's peaks. Second, on the 200-m scenes the oracle itself stays about 3 dB above the reference: the oracle removes ΔC_0 exactly, and what is left comes from the linear and quadratic residuals ΔC_1, ΔC_2, which this correction does not address. Third, the spike scene is the resolution limit: a 500 m jump between scatterers 200 m apart ($< \delta x$) cannot be separated in the sub-aperture spectra, and the interpolated block is wrong for both. One remark on the 3-target scene: its outer targets are 6 km apart, inside the collision band of Section 6.3, but both are at $h = 0$, so the collision has no effect there; T17b exposes it.

6.3 The Doppler-alias collision

T17b moves two targets (240 / −120 m) apart and records the ambiguity against their separation d (Fig. 15). Outside two narrow bands SATA equals the oracle; inside $|d - X| \lesssim 0.6$ km and $|d - 2X| \lesssim 0.6$ km it does not reach it. T17a, six scenes of three targets at random positions 3.8–4.9 km apart with random heights in $[-300, 400]$ m and random baselines, is within 0.33 dB of the oracle.

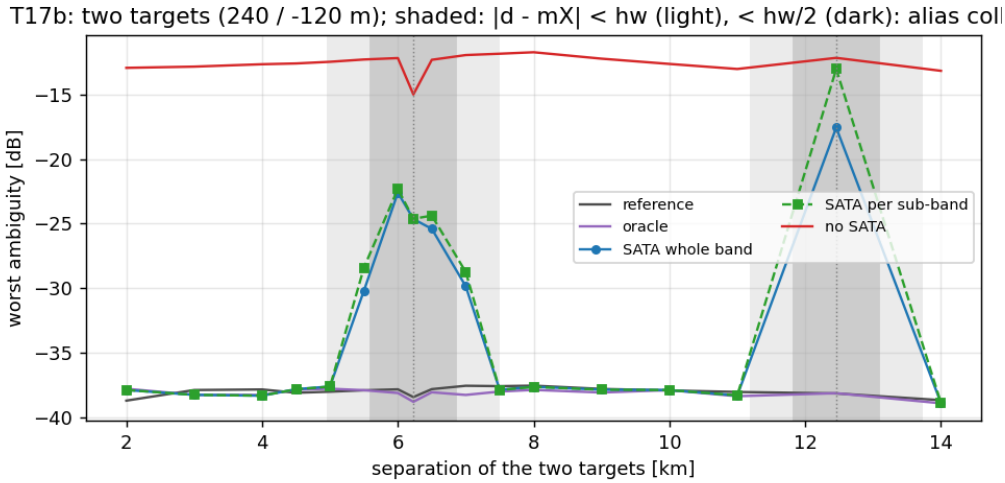


Figure 15: T17b: two targets (240 / −120 m) against their separation. Shaded: $|d - mX| < h_w$ (light) and $< h_w/2$ (dark, the first null of the window response). Outside the dark core SATA equals the oracle; inside, the two main lobes share the same bins and no per-channel correction can serve both.

Mechanism. By Eq. (5), two scatterers $d = X$ apart have, in every window that sees both, true Dopplers that differ by exactly $\mathrm{PRF}_{\mathrm{ch}}$. On the channel line, sampled at $\mathrm{PRF}_{\mathrm{ch}}$, those two

Dopplers fall into the *same bin* (Fig. 16). The bin then holds the direct component of one scatterer and the folded component of the other, with different channel-phase vectors $\varphi_i(A)$ and $\varphi_i(C)$, and one scalar per channel cannot make both consistent with the model. The map can only choose one value; the other component then leaks into that output band with the *difference* pattern $\varphi_i(C) - \varphi_i(A)$. For the targets of Fig. 15 that pattern is $(118°, 50°, 8°, 3°)$, a spread of 0.32 cycles, larger than the spread of either target alone (0.21 and 0.11 cycles). This is a property of any correction applied per channel, bin by bin, on the aliased line, before the sub-bands are unmixed: the oracle escapes it only because it acts on the separated signals, which are not available. The core of the collision is set by the main-lobe width of the window response, $\pm 2\mathrm{PRF}_{\mathrm{ch}}/T_{\mathrm{sub,eff}} = \pm 50\,\mathrm{Hz}$, i.e. ± 41 cells $= \pm 0.63\,\mathrm{km}$ around mX: as long as $|d - mX|$ exceeds it, the two components sit in different bins and can be corrected separately.

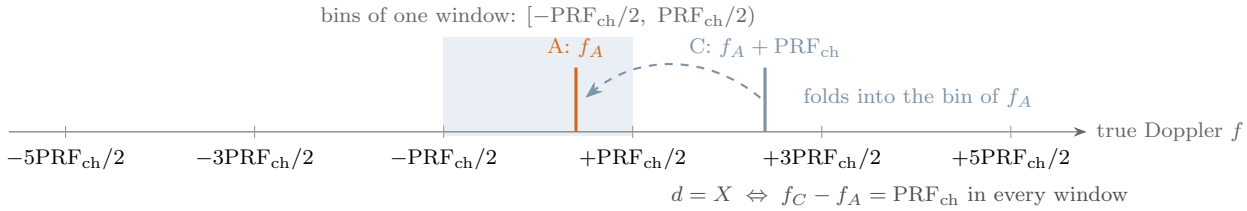


Figure 16: The collision. The sub-aperture spectrum of the channel line only has bins for $[-\mathrm{PRF}_{\mathrm{ch}}/2, \mathrm{PRF}_{\mathrm{ch}}/2)$; a component at $f_A + \mathrm{PRF}_{\mathrm{ch}}$ lands in the bin of f_A. Two scatterers X apart have exactly that Doppler difference in every window, so their energies coincide bin by bin.

What the test found in the map, and the repair. A first run of T17b showed the collision band twice as wide, $|d - X| < h_w$, with the ambiguity at the uncorrected level throughout it (e.g. $-18.9\,\mathrm{dB}$ at $d = 5.0\,\mathrm{km}$ against an oracle at -37.8). At those separations the two components are in *different* bins ($|d - X| = 1.2\,\mathrm{km}$ is 100 Hz apart, twice the main-lobe width) and can be corrected separately, so the loss was not inherent. Its cause was the rule then used to draw the map, "where a folded image meets the direct block of another scatterer, the direct one wins the whole overlap": the direct block of C extends $h_w = 82$ cells on each side of x_C, twice the main lobe, and overwrote the folded main lobe of A wherever it fell inside that block, giving A's folded energy the phase of C in every window. The rule was replaced by the nearest-image rule of Section 4 (A5), which splits the overlap at the midpoint between the two components. The change affects only overlapping images: for a single scatterer, for the 3-km scenes and for every scene of the sweep (spans $\leq 2.4\,\mathrm{km}$) the map is bit-identical. With it, $d = 5.0$ and $7.5\,\mathrm{km}$ return to the oracle, the transition narrows to the main-lobe width, and inside the core the result improves from -14 to $-22 \ldots -25\,\mathrm{dB}$: still 13–16 dB from the oracle, which is the inherent part, but 10–13 dB better than no correction.

Where 1D SATA on the channel can be trusted. When the residual is similar at azimuth positions one fold X apart at the same range (6.2 km in this geometry), or when scatterers with different residuals are not found at that spacing. For the scenes of the case matrix (spans $\leq 2.4\,\mathrm{km}$, all inside one fold) the condition is respected. For continuous terrain the condition is that the height, relative to the reference surface, does not change strongly over X; how strongly is "strongly" is given by the closed form, Eq. (2): with $b_{xt} = 100\,\mathrm{m}$, a height difference of 100 m over X is a phase difference of 52° on that channel.

7 Results

7.1 One target

Figure 17 shows the impulse response of one target at 240 and 400 m, with the DPCA array and $b_{xt} \sim U(0, 100)$ m. The ambiguous replicas at multiples of $1/\mathrm{PRF}_{\mathrm{ch}}$ in azimuth time (± 0.81, ± 1.62 s) rise to -13 and -9 dB without correction and disappear with SATA, whole band or per sub-band, to -39.4 dB.

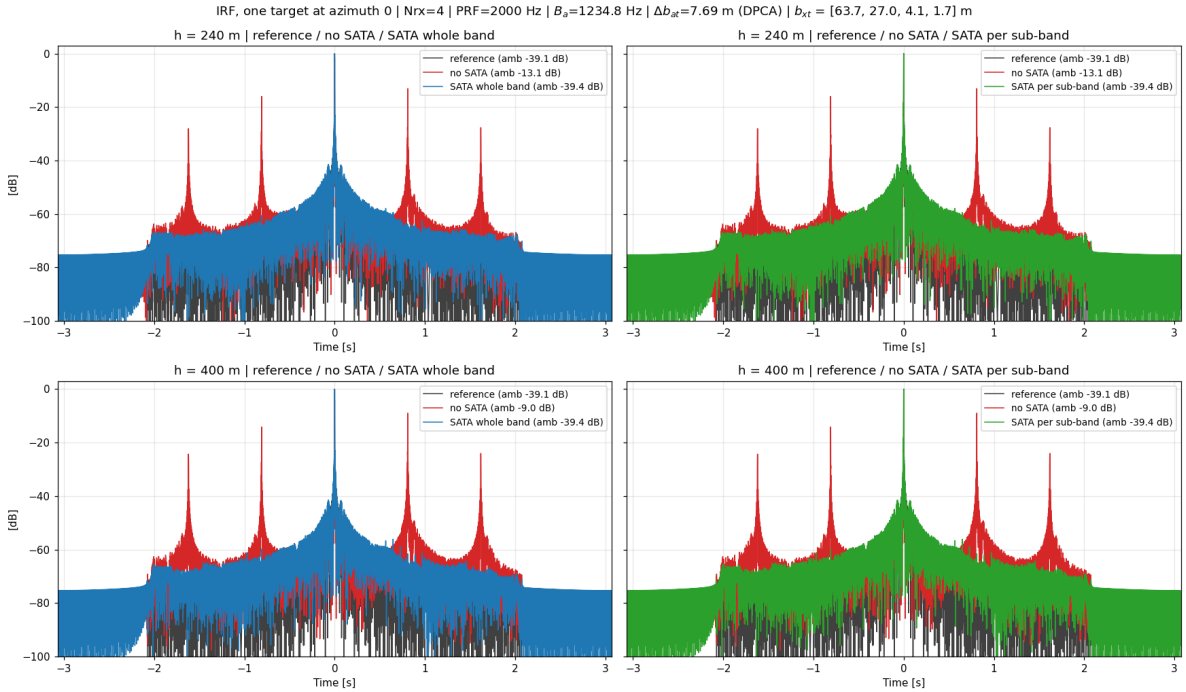


Figure 17: Impulse response of one target, $h = 240$ m (top) and 400 m (bottom); left whole band, right per sub-band (`plot_irf_single_target.py`). The broad floor between -75 and -60 dB is the 2 % residual of T9 spread by the azimuth compression; it lies well below the ambiguity level.

7.2 The geometry sweep

The sweep repeats the case matrix of the report with the current code: 181 scenes in nine families of height profiles, 2 to 13 targets 200 m apart on the iso-range surface, $b_{xt} \sim U(0, 100)$ m; plus the topographic ramp of the reference scenes (18 scenes, Section 7.3). Every scene is reconstructed with: reference (filters told the true heights), no correction, SATA whole band, SATA per sub-band, the previous map in both variants, and the oracle. With $b_{xt} = 0$ every variant is identical to no correction, as it should be.

What the spacing implies. The spacing of 200 m is below $\delta x \simeq 309$ m and far below $2h_w$. All footprints touch and every scene is a single terrain block in which the map interpolates linearly between the targets. The sweep therefore tests that interpolation. No scene spans more than 2.4 km, so there are no alias collisions.

Table 5: Median worst ambiguity [dB] per family (181 scenes, $b_{xt}^{\max} = 100\,\mathrm{m}$).

family	scenes	ref.	no SATA	whole	sub-band	prev. (whole)	prev. (sub)
ramp	78	−34.5	−22.5	−31.0	−31.0	−28.2	−23.3
zigzag	26	−34.1	−30.8	−30.8	−30.8	−30.8	−30.8
piecewise	15	−33.9	−14.1	−28.3	−28.2	−19.7	−15.7
steepening	8	−34.1	−12.8	−30.8	−30.8	−21.0	−15.7
flattening	8	−34.1	−12.7	−30.8	−30.8	−20.3	−15.7
cliff	14	−34.1	−12.7	−25.6	−25.6	−19.8	−17.2
spike	8	−34.1	−8.3	−11.9	−11.9	−10.0	−8.4
mesa	6	−34.1	−8.0	−18.7	−18.7	−11.3	−8.2
random heights	18	−33.9	−7.9	−17.8	−17.8	−15.4	−13.4

Overall. Against the previous map the new one is better by more than 0.5 dB in 125 of the 181 scenes, equal in 53 and worse in 3 (cliffs of 1 km and random heights up to 1 km, where neither map represents the scene); the median improvement is 3.7 dB (Fig. 18). Whole band and per sub-band differ by at most 0.07 dB in every scene. Figure 19 orders all scenes by the largest height difference between neighbouring targets: with the new map every scene with jumps up to about 100 m stays at the oracle level (3.3 dB above the reference), whatever the family or the number of targets; the error grows only beyond that, where one interpolated value cannot serve two targets 200 m apart. The previous map degrades from jumps of about 10 m.

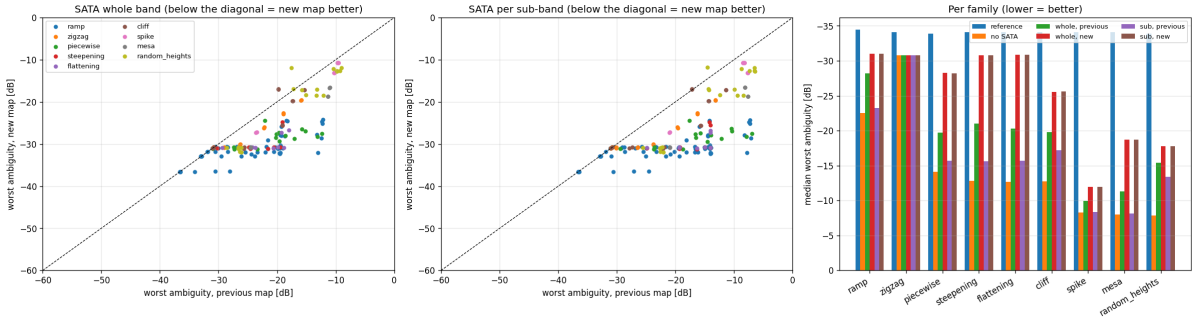


Figure 18: Sweep overview. Left and centre: new map against the previous one, one dot per scene; below the diagonal the new map is better. Right: median per family.

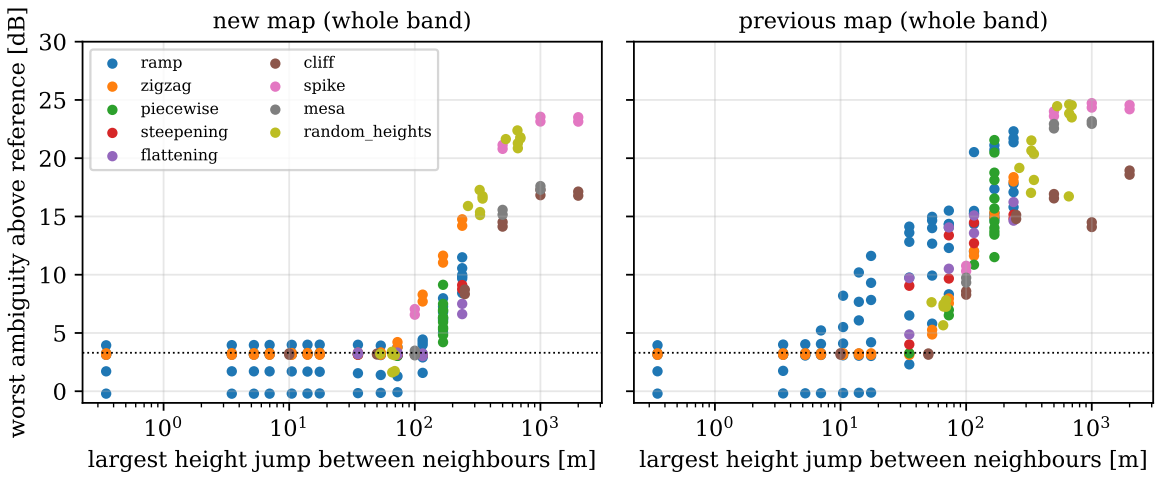


Figure 19: Excess ambiguity over the reference against the largest neighbour-to-neighbour height jump, all 181 scenes, whole band. Dotted: the oracle level of the 200-m scenes. Left: new map. Right: previous map.

Improvement over no correction, scene by scene. Figures 20 and 21 give the gain of SATA over the uncorrected reconstruction (worst ambiguity without minus with SATA; positive: SATA better), next to the height profiles of each family. For the uniform ramps the gain grows with the inclination and the number of targets up to 26 dB, median 10.1 dB against 5.8 dB for the previous map, and falls again at 40–50°, where neighbours are 170–240 m apart in height. A gain near zero is not a failure: in all 36 ramp and zigzag scenes with less than 0.5 dB of gain the uncorrected reconstruction is already within 4 dB of the reference, and there is nothing left for a ΔC_0 correction to remove. The smooth profiles (steepening, flattening, piecewise) gain 7–23 dB; spike, mesa and random heights gain 2–11 dB, limited by their jumps. No scene gets worse with SATA (smallest gain −0.03 dB).

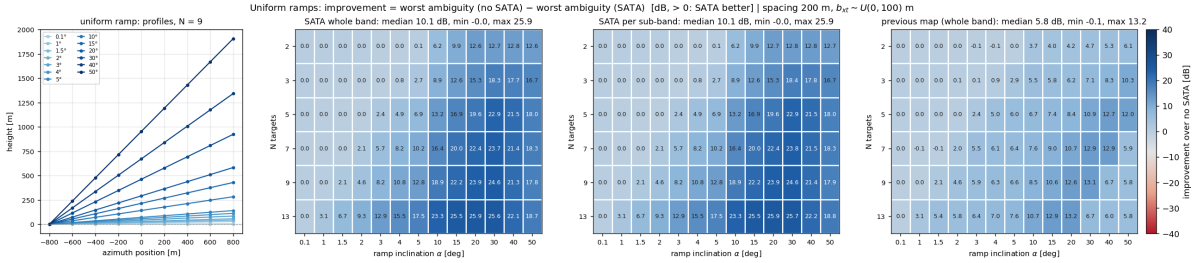


Figure 20: Uniform ramps. Left: height profiles for $N = 9$, one curve per inclination. Right: improvement over no correction [dB], N targets (rows) against inclination (columns); SATA whole band, per sub-band and the previous map.

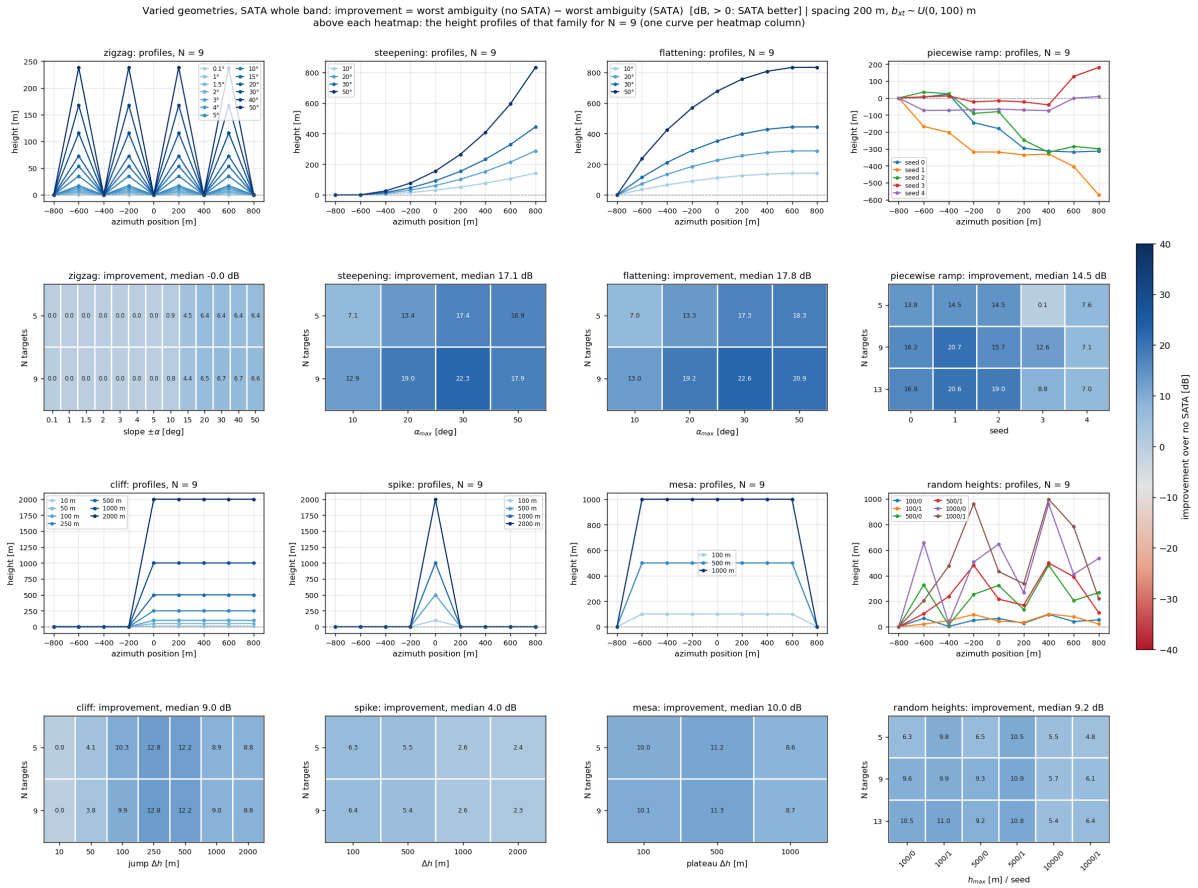


Figure 21: Other height profiles, SATA whole band. For each family, top: its height profiles for $N = 9$, one curve per heatmap column; bottom: improvement over no correction [dB].

Selected scenes. Figures 22–25 show one scene per regime, in the layout of the sweep script: top left the impulse response of the reference and of SATA (both variants), bottom left reference and no correction, right the target profile; the titles carry the worst ambiguities, including the previous map and the oracle.

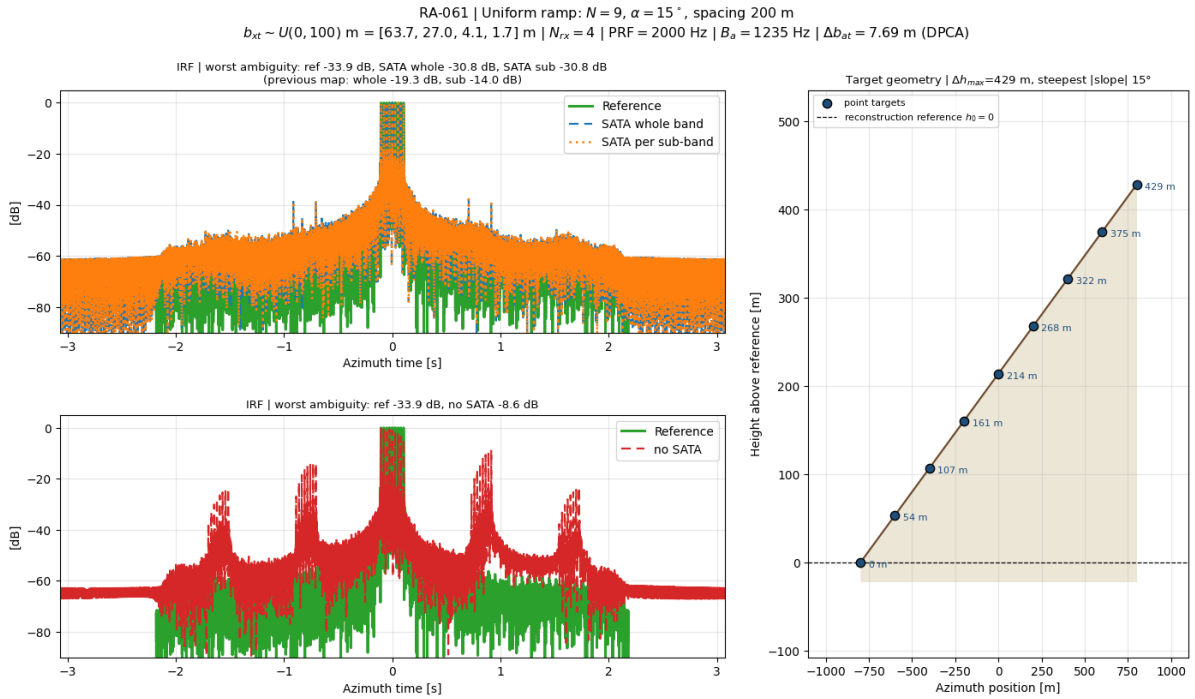


Figure 22: RA-061, ramp of nine targets at 15° (54 m between neighbours). Smooth terrain: the interpolated block is the correct residual everywhere, and SATA brings the ambiguity from −8.6 to −30.8 dB, the oracle level; the previous map stops at −19.3 dB.

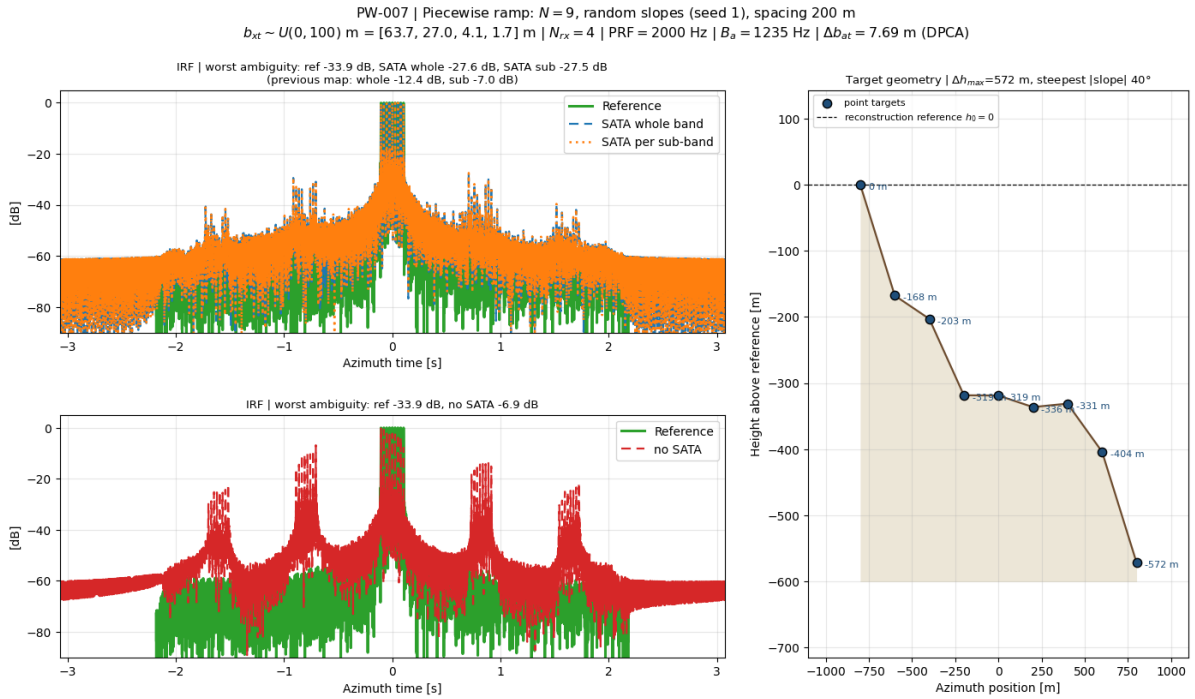


Figure 23: PW-007, piecewise ramp with random slopes up to 40° (jumps up to 168 m). The kinks cost a few decibels against the smooth ramp: −27.6 dB with SATA, −12.4 dB with the previous map, −6.9 dB without correction.

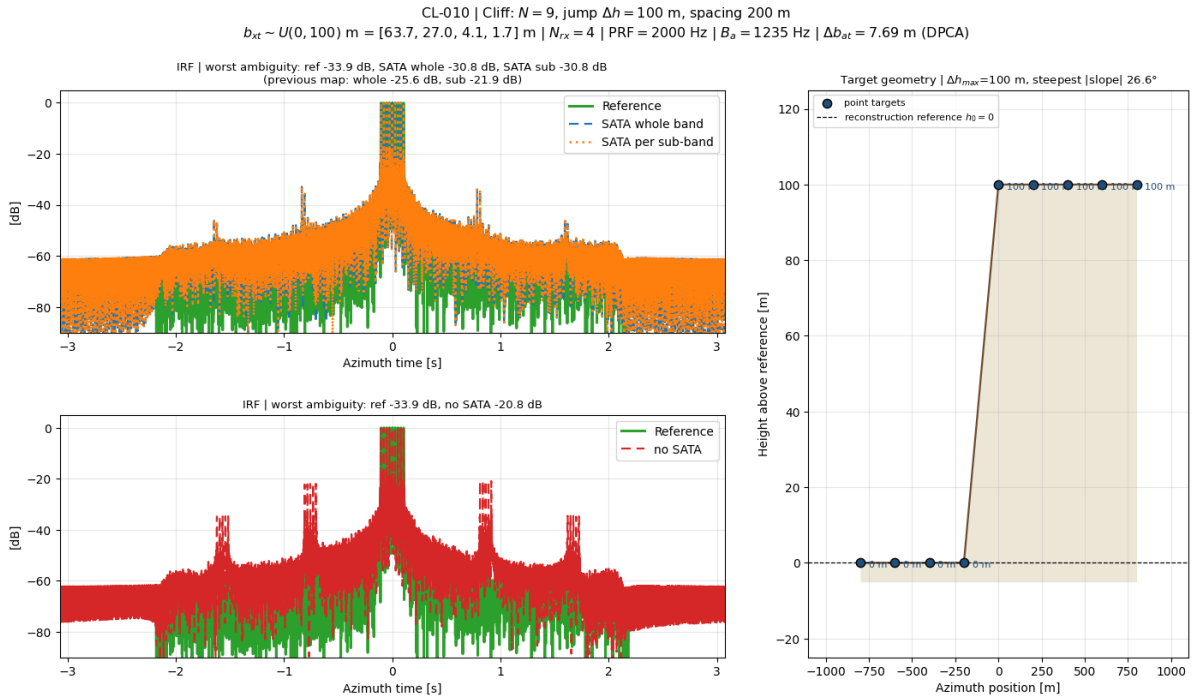


Figure 24: CL-010, cliff of 100 m in a line of nine targets. A single jump of 100 m is at the limit of Fig. 19: SATA reaches −30.8 dB (oracle level), the previous map −25.6 dB.

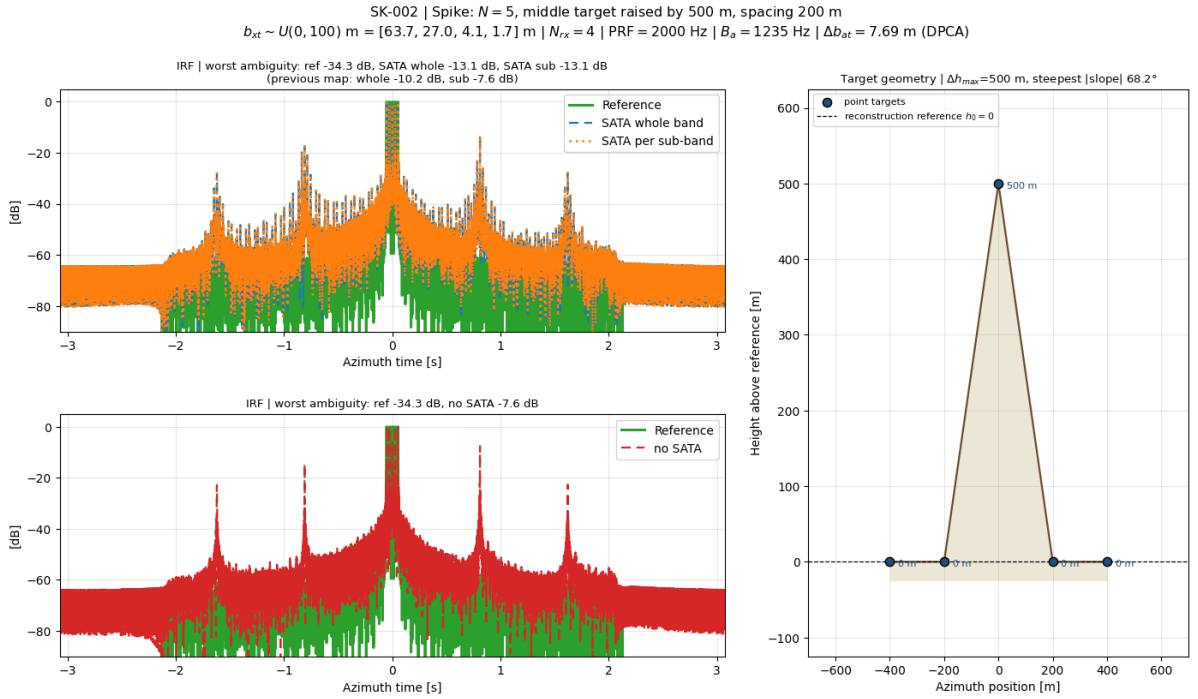


Figure 25: SK-002, one target 500 m above four at zero height, 200 m apart (the spike scene of T11). The central target's residual cannot be separated from its neighbours in any sub-aperture spectrum; SATA reaches −13.1 dB, the oracle −31.0 dB.

7.3 The topographic ramp

The reference scenes of the report use a topographic ramp: N targets at the *same* azimuth on the line $\Delta h = \Delta y \tan\alpha$ through the scene centre, Δy from 10 to 100 % of 2 km in ground range. It is a family of the sweep (`topo_ramp`, $\alpha \in \{5, 10, 14.04, 15, 20, 30\}°$, $N \in \{5, 9, 13\}$). Its residuals include the flat-earth term and span a wide interval (-1.6 to -16.0 cm on channel 0 for $N = 9$, $\alpha = 14.04°$). The 1D simulation sums all targets into one azimuth line, where they occupy a single azimuth cell, and no map on that line can give them different values (Fig. 26): summed in one line, SATA gains only 2.3 dB (median), exactly like the previous map. In a real acquisition these targets are separated in slant range (18–520 m from the centre, at least 13 m between neighbours, i.e. two range cells at 6 m resolution) and each range line is corrected with its own map (report, Section 4.3). Correcting each target on its own line gives a median gain of 31.0 dB and reaches the oracle to within 0.03 dB in all 18 scenes (Fig. 27). The exception is $\alpha = 20°$, equal to the look angle: the ramp then follows the iso-range line, all targets fall within 3 m of slant range and nothing separates them. The topographic ramp is thus a test of the range dimension, which the per-range-bin (2D) processing handles, not of SATA 1D on one line.

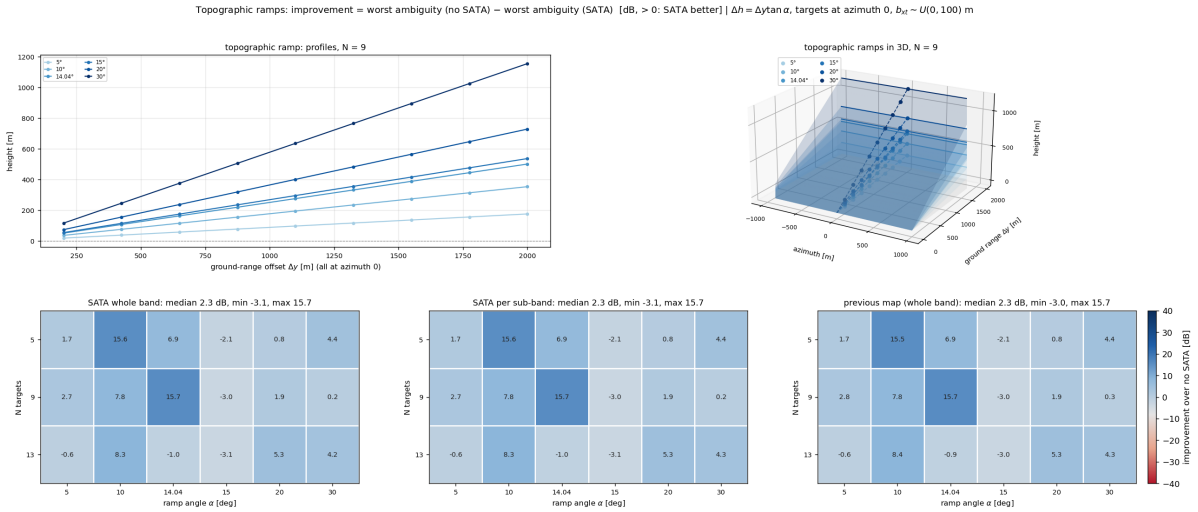


Figure 26: Topographic ramps. Top: the ramps for $N = 9$, as height against ground-range offset and in 3D (targets as dots at azimuth 0; the surfaces extend the ramp along azimuth only to show its slope). Bottom: improvement over no correction [dB], N targets against ramp angle; SATA whole band, per sub-band and the previous map.

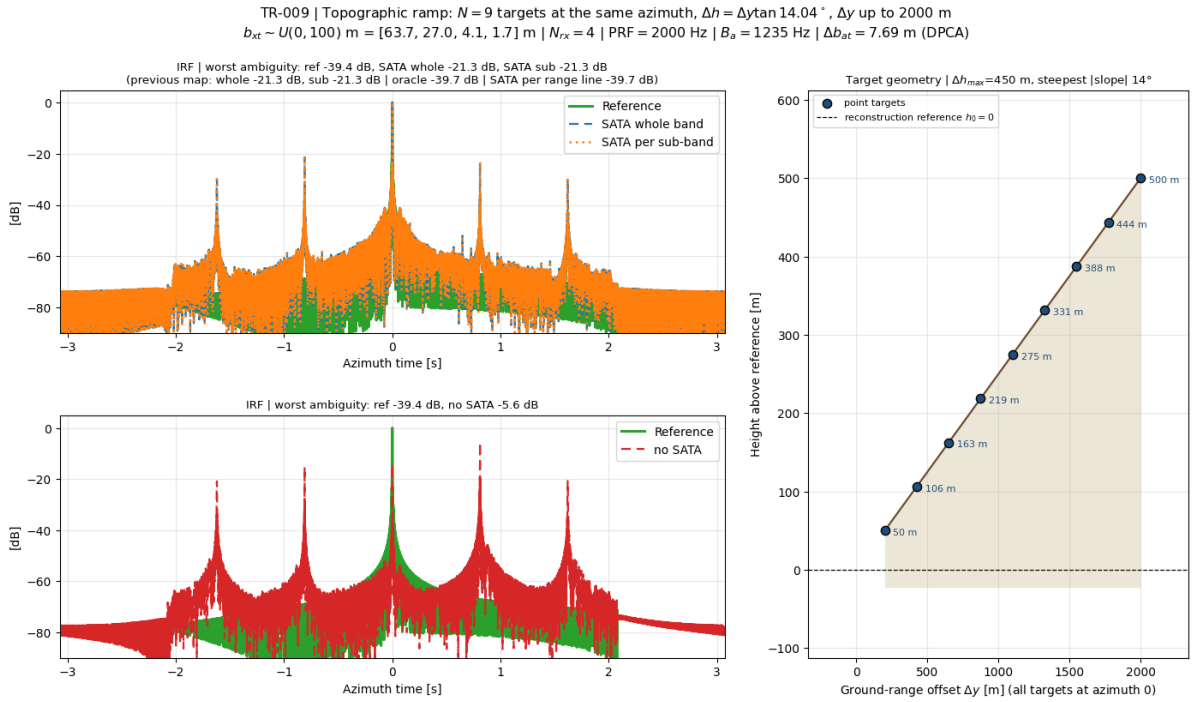


Figure 27: TR-009, the reference topographic ramp of the report ($N = 9$, $\alpha = 14.04°$). On one summed line SATA stops at -21.3 dB; per range line it reaches -39.7 dB, the oracle, against -39.4 dB for the reference.

8 Conclusions

The SATA 1D correction of ΔC_0 is implemented as the report describes, and every convention it rests on was traced through the code and confirmed by a test: the kernel reproduces its input for a null map and applies an exact global rotation for a constant one; only the removal sign brings the reconstruction to the reference; the ruler places every STFT peak on its scatterer's cell or on a predictable fold image $\pm X$; the map holds each scatterer's residual where its energy lands, so that with two scatterers in one window each reads its own value. For one target the ambiguities return to the reference level at every height. For several targets the correction matches the oracle whenever the scatterers are resolvable and not in alias collision; on the 181 scenes of the sweep the interpolated map holds the oracle level as long as neighbouring targets differ by less than about 100 m, and beats the previous map by a median of 3.7 dB.

Two things were found and are documented. The per-sub-band kernel needed its frequency axis replaced by the absolute Doppler of the band on the broadside grid; the previous axis made it behave as the whole-band kernel, and for the iso-range scenes simulated here both give the same result. And the Doppler-alias collision: two scatterers at the same range whose azimuth separation is a multiple of $X = 6.2$ km, to within the main-lobe width of the sub-aperture response (0.6 km), share the same sub-aperture bins in every window, and no per-channel, bin-wise correction can serve both. It is a property of correcting before the unmixing, not of the implementation, and it bounds where 1D SATA on the channel can be trusted: where the residual is similar at positions one fold apart. Testing it also exposed and repaired a weakness of the map (the overlap of images is now split between the nearest components), which had widened the affected band to $\pm h_w$ and cost up to 19 dB at separations of $X \pm 1.2$ km.

Three further limits belong to the method: scatterers closer than $\delta x \simeq 309$ m share one correction; the linear and quadratic residuals ΔC_1, ΔC_2 are not corrected, which accounts for the 3 dB between oracle and reference on dense scenes; and targets displaced in range (the

topographic ramp) must be corrected on their own range lines, which is what the 2D processing does. Beyond the simulation, the map requires the scatterer heights, which in practice come from a DEM.

A Running the Code

file	content
`sata1d/config.py`	system, scene, array; `make_config` (DPCA b_{at}, $b_{xt} \sim U(0, b_{xt}^{\max})$)
`sata1d/geometry.py`	transmitter and receiver tracks
`sata1d/signal_model.py`	point-target raw data, reference and channels
`sata1d/reconstruction.py`	`GetCoeffNu`, inversion filters, reconstruction
`sata1d/sata.py`	Stage A (map) and Stage B (`sata_1d`), whole band
`sata1d/subband.py`	per-sub-band fit, axis, kernel and reconstruction
`verify_sata1d.py`	tests T1–T19 of Section 6 (writes `verification/`)
`plot_irf_single_target.py`	Fig. 17
`plot_sata_windows_single_target.py`	window-by-window PDF of the kernel
`run_geometry_sweep.py`	geometry sweep and topographic ramp (Section 7.2)
`plot_sweep_improvement_heatmaps.py`	Figs. 20–26 from the sweep CSV

```
pip install -r requirements.txt
python verify_sata1d.py                                    # ~15 min, writes verification/
python plot_irf_single_target.py
python run_geometry_sweep.py --bxt 100 --workers 4         # 181 profile scenes, ~5 min
python run_geometry_sweep.py --families topo_ramp --bxt 100
python plot_sweep_improvement_heatmaps.py geometry_sweep/sweep_results.csv
```

Minimal use of the package on a scene of one's own:

```
import sata1d as S
cfg = S.make_config([(-1500.0, 240.0), (1500.0, -120.0)])   # (x [m], h [m]) per target
tr  = S.build_platform_tracks(cfg)
ch  = S.generate_channels(cfg, tr)
img_none  = S.reconstruct(cfg, tr, ch)
img_whole = S.reconstruct(cfg, tr, S.sata_channels(cfg, tr, ch))
img_sub   = S.reconstruct_subband(cfg, tr, ch)
```